ADHD planner: product design and research

Sep 28, 2026 · @Devesh

1. Executive summary

Build an Apple-native planner that learns each person's own medication focus window from one-tap check-ins and fills it with the work they choose. It should also survive bad days without shame, and let them flip their whole learned routine to "Lifestyle 2" for the holidays and back in one tap. Nothing on the market does this: medication trackers (FocusCurve, Pillr, Theraview) draw curves but schedule nothing, and planners and AI schedulers (Tiimo, Structured, Motion, Reclaim) schedule everything but ignore medication.

The core loop. Log a dose (Watch, widget, Siri, Action button) → the app shows your usual focus window, learned from your own check-ins → the solver puts your chosen deep work inside it and body, social and admin tasks outside it → outcomes feed a behaviour ledger → nightly on-device learning refines durations, timing and the window.

What the research changed in the original design

- **Label durations are only a starting point.** 57% of 1,327 lisdexamfetamine users report 7 hours or less against a label showing effect to 14. The fixed "T to T+9h" rule becomes a personal, learned, confidence-labelled window (sections 6 and 7).
- **Planning is the biggest deficit, not memory** (d = 1.60 for planning; recall of a made plan was normal). The app should do the planning and make time visible at the point of performance: Watch, Lock Screen, Dynamic Island (sections 3 and 4).
- **Lapses are the default.** Across mental-health apps only 3.3% of users remain at day 30, and emotional dysregulation is large in ADHD (g = 1.17). Amnesty, a baseline day, forgiving progress and 30-second re-entry are core features, not settings (section 5).
- **One red line.** Any suggestion about when or how much to take, boosters included, would make the app a regulated medical device in Australia, the US, the UK and the EU. The app logs, learns a self-reported "focus pattern" and moves tasks; it never moves doses (section 8).
- **Apple sets hard limits.** Apps can read but not write Health doses; there's no API for the Sleep Schedule; iOS gives no reliable background time; App Review bars personal health information in iCloud. Each has a workaround (section 11).
- **Learning is small, local Bayesian arithmetic,** not a black box. Every learned number shows its evidence; standing changes need the user's approval (section 9).

Lifestyle 2. Each lifestyle keeps its own learned routine; the dormant one is frozen, so switching back is instant. A new Lifestyle 2 starts from the user's own weekend pattern and three quick answers, and is about 70% personalised after a week (section 10).

Biggest risks. Drift-off after two weeks, a scheduler that over-packs the day or makes one bad move, setup burden, and regulatory or reputational exposure around stimulants. Mitigations are in section 13.

Next steps. Form a Pty Ltd, get a health-regulatory lawyer's review of the red lines and wording, and run two weeks of technical spikes before building the MVP (section 17). The ten decisions in section 18 are yours to make.

2. Scope, assumptions and generalisation

The original inventory encodes one person's regime; the product must turn every personal rule into a configurable policy with sensible defaults. That regime then becomes the first profile template and the main test fixture.

Assumptions used throughout

- Users: adults and older teens with diagnosed or suspected ADHD, on or off medication. Carers of younger children are out of scope for v1.
- Platforms: iPhone first, then Apple Watch, Mac and iPad. No Android at launch.
- Market: Australia first (PBS medicines, Schedule 8 rules, TGA, Australian Privacy Principles), then US, UK and NZ.
- Positioning: a planning and self-management tool. It records what the user and their prescriber decided; it never recommends a dose, a timing change or a booster.
- Canonical store: the app owns its own data model and mirrors into Apple Calendar and Reminders. Apple's stores are outputs and inputs, not the source of truth.

How the personal rules generalise

Rule in the original inventory	Generalised policy object	Default for a new user
Productive window = T to T+9h after medication	`EffectWindow` per medication: onset, peak, plateau end, wear-off, each learned per person	Label-range starting estimate, user-editable, refined by check-ins
Lunch no earlier than T+5h	`MealPolicy` anchored to a medication event, a clock time, or appetite check-ins	Anchored to expected appetite return; eat-something prompt if nothing logged

Rule in the original inventory	Generalised policy object	Default for a new user
Work-day lunch = max(T+5h, home time)	Composite anchor: latest of several constraints	Same structure, user picks the constraints
Work and exam prep > uni > social > gym > meals > errands	`PriorityLadder`, reorderable per Lifestyle and Mode	Commitments > deadlines > health basics > chosen goals > errands
Gym at 11am or 3pm, skipped on work, social and exam days	`MovementPolicy` with candidate slots and skip conditions	Up to three candidate slots; skip conditions offered as toggles
Exam period 9 to 27 November	Dated `Mode` overlay (Exam, Holiday, Sick, Travel)	Offered when assessments are imported
Dinner 8:30pm, 11pm reflection, watch charging, sleep	`Routine` anchors, each tagged calendar-worthy or checklist-only	Morning and evening checklists only; nothing trivial on the calendar
Reminder and Calendar mirroring rules	`SyncPolicy` per list and calendar	Paired by internal ID, never by title
Norwest gym and nearby errands	`SavedPlace` with opening-hours source and learned travel overhead	Home, work or school, one gym or shop

Terms used in this document

- **Effect window**: the internal name for the period after a logged dose when this user reports they can do demanding work. Personal, learned from their check-ins, never prescribed by the app. Users only ever see it called their "focus window" (see section 8 for why).
- **Booster**: a second, usually short-acting dose the prescriber has already authorised. The app schedules around it; it never suggests one.
- **Lifestyle**: a learned behavioural baseline (term time, holidays, shift work). The Lifestyle 2 toggle switches which baseline the planner uses.
- **Mode**: a temporary rule overlay (Exam, Sick Day) that sits on top of whichever Lifestyle is active.
- **Hard rule**: a constraint only the user can change. **Learned preference**: a pattern the app found and the user approved.

3. The ADHD problem space

The biggest measured deficits are planning (d = 1.60) and time perception, not remembering a plan once it exists. So the app's job is to do the planning, make time visible, and survive the user's bad days. The same person also carries large emotional dysregulation (g = 1.17), circadian delay in about 3 in 4 cases, and a medication schedule that none of their current tools understand.

3.1 Struggles by domain

Domain	What it looks like day to day	Evidence (strength)	What current tools miss
Time blindness	Late, misjudged durations, plans built on rare hyperfocus days	Deficits in every timing paradigm, 55 studies (Marx et al. 2021) (strong)	Reminders fire once at a clock time; nothing shows elapsed or remaining time
Planning and sequencing	Can't turn a goal into ordered steps; freezes at "where do I start"	Planning d = 1.60; recall of a made plan was normal, ~86.5% in both groups (Fuermaier 2013) (moderate, n = 90)	To-do apps assume the user can plan, the single biggest deficit
Task switching and transitions	Can't leave one task for the next; overruns	Switching d = 0.94 (same study)	Calendars jump from block to block with no ramp
Initiation and procrastination	"ADHD paralysis"; deadline-driven crisis work	r = 0.42 with symptoms, n = 490 (Oguchi 2021)	Tasks sit as monolithic items with no first step
Hyperfocus	Loses hours, misses meals, blows past the next commitment	Separable feature, two samples n = 251 and 372 (Hupfeld 2019)	No planner interrupts; they only prompt
Waiting mode	Can't start anything before an appointment; the whole morning is lost	Widely reported, no clinical research (Healthline) (anecdotal)	Nothing makes the gap before an event feel safe to use

Domain	What it looks like day to day	Evidence (strength)	What current tools miss
Emotional dysregulation and shame	One missed plan becomes a spiral and app abandonment	g = 1.17, 13 studies, ~70% of adult patients report it (Beheshti 2020) (strong). RSD: 5 small qualitative studies, not a diagnosis (review)	Red overdue counts, broken streaks, piles of missed notifications
Boredom and novelty	Apps feel dead after 2 to 3 weeks	r = 0.40, 18 studies, >22,000 people (Muris 2026)	Static UI, same prompts forever
Sleep	Late chronotype, revenge bedtime procrastination, late dose reaching bedtime	73 to 78% delayed sleep/wake; melatonin onset ~90 min late in adults (Luu & Fabiano 2025); lisdexamfetamine insomnia 11 to 19% vs 4 to 5% placebo (Najib 2017)	Dose reminders at a fixed clock time; sleep and planner apps don't talk
Eating and hydration	Skips lunch on meds, ravenous at wear-off, dry mouth	Decreased appetite 26.5 to 33% vs 2 to 6%; dry mouth 26 to 32% (Najib 2017); weaker hunger sensing (Göz Tebrizcik 2025)	Food, water and meds live in separate apps
Medication adherence	Forgets doses, can't remember if already taken, runs out	Nonadherence 13 to 64%, forgetting the top unintentional reason (McCarthy 2014); fewer than half of adults "consistently medicated" (ADDitude)	Pill apps log "taken" and stop there: no effect window, no day plan

Domain	What it looks like day to day	Evidence (strength)	What current tools miss
Rebound and wear-off	Irritable, teary, crashes mid-to-late afternoon for ~1 hour	Described as common with short-acting stimulants; adult prevalence unmeasured (ADDitude) (weak)	No tool warns ahead or moves hard tasks away from it
Supply and shortages	Calling pharmacy after pharmacy; rationing	UK: 97% affected, 70% rationed (ADHD UK); US: 38% (ADDitude); Australia short 2023 to 2025, prescriptions up 60 to 140% (The Conversation)	Nothing tracks repeats, script expiry, or stock calls
Menstrual cycle and perimenopause	Meds feel weaker some weeks; symptoms surge in the 40s	Worst symptoms at menstruation, doses unchanged (n = 30, Zaritsky 2026); 94% worse in perimenopause (self-selected n ≈ 1,500, ADDitude)	Cycle and medication apps never cross-reference
Money	Late fees, forgotten subscriptions, impulse buys	62% impulsive shopping, 57% miss loan payments (secondary, ADDitude) (weak)	Banks alert after the charge
Household and admin	Doom piles, paperwork, lost keys	Clinically significant hoarding 20% vs 2% (The Conversation)	Chore lists go invisible once the app closes
Social upkeep	Texts left on read, birthdays missed, lateness	75% feared relationship harm when unmedicated (ADHD UK)	No cross-app reply-later queue
Digital distraction	Opens the planner, ends up in a feed	Social-media disorder 15% vs 3.3% (OR 5.25), escape motive (Thorell 2025)	The planner lives on the distraction device

Domain	What it looks like day to day	Evidence (strength)	What current tools miss
Comorbidity	Anxiety, low mood, autism, dyslexia change what "good design" means	Up to 80% have another disorder; anxiety ~50% (Katzman 2017); 27% of autistic adults have ADHD (Yerys 2025)	Loud, stimulating "ADHD" apps overwhelm anxious or autistic users
Context shifts	Routines collapse in holidays, weekends, travel, shift changes	No direct research; mechanism is loss of external anchors plus circadian drift	One weekly template for the whole year

3.2 What the pattern means for the product

- **Visibility beats storage.** 61% of ADDitude readers prefer paper because it stays in view; one reader said a digital calendar "ceases to exist for my brain once closed" (ADDitude). Widgets, the Watch face and Live Activities are the product; opening the app is the exception.
- **Lapses are the default path.** Across 93 mental-health apps, day-15 retention was 3.9% and day-30 3.3% (Baumel 2019). Coming back after two weeks away must take under 30 seconds and carry no backlog guilt.
- **Medication, sleep, food and focus are one coupled system.** A late wake shifts the dose, the dose shifts the productive window, the window shifts lunch and bedtime. No mainstream tool models that chain.
- **Support, not treatment.** Digital ADHD interventions show small pooled effects ($g = -0.32$, 23 RCTs, mostly children, Kasturiratna 2025). Sell daily function and better prescriber conversations, never symptom reduction.
- **Australia has its own admin burden.** About 800,000 Australians have ADHD at a cost above A$20 billion a year; the 2023 Senate inquiry made 15 recommendations (The Conversation). NSW trained GPs could continue stimulant prescriptions from 1 September 2025, and state rules still differ at borders.

Research gaps worth knowing: no study measures waiting mode, ADHD burnout prevalence, holiday routine collapse, or ADHD with shift work. The app's own data could become the first evidence on some of these.

4. Evidence-based solutions and what they mean for design

The core should copy the active parts of adult ADHD CBT: calendar plus task list, task breakdown, distraction capture, reframing, self-reward and weekly review. It should deliver them at the moment they're needed (Barkley's "point of performance"). Most popular app tricks (streaks, points, body doubling, Pomodoro, dopamine menus) have weak or no ADHD-specific evidence, so they ship as optional layers, never the spine.

4.1 Techniques graded

Technique	Evidence	Key numbers	How the app uses it
CBT / meta-cognitive therapy skills	Strong; guideline-endorsed (AADPA)	Safren: responders 67% vs 33%, gains held at 12 months (JAMA); Solanto: 42% vs 12% (AJP); Cochrane SMD −0.84 vs waitlist (NIHR)	Skill spine: capture → break down → schedule → park distractions → review and reframe
Structured student programme (ACCESS)	Moderate	n = 250 students, inattention d = .50, executive function d = .56 (RCT)	Semester-shaped onboarding and weekly check-in for uni users
Externalising time	Theory strong, device efficacy weak	Timing deficits medium; delay aversion small-to-medium, real rewards double the pull (Marx 2021)	Shrinking time rings, leave-by countdowns, 10/5/1-minute transition haptics
Implementation intentions (if-then)	Moderate	d = 0.65 general; guided plans g = 0.47 vs written g = 0.28 (MCII meta); normalised inhibition in ADHD children (Gawrilow 2008)	Every scheduled task gets "When I [cue], I will [first action]", with the cue wired to a trigger

Technique	Evidence	Key numbers	How the app uses it
Acute exercise before study	Moderate	Inhibition g = 0.55 in ADHD adults (2026 meta)	Optional 10 to 20-minute "movement primer" before deep work; gym placed before study where possible
Notification batching	Moderate (general population)	3 batches a day: inattention d = −0.65; no notifications raised anxiety d = 0.56 (Fitz 2019)	Daily notification budget; non-urgent items in 2 to 3 digests
Receptivity-timed prompts (JITAI)	Moderate (general)	Prompt effect +66% on day 1, fading ~2% a day to zero by day 28 (HeartSteps); receptivity models +36% response (Mishra & Künzler 2021)	Learned send times, a "send nothing" option, rotated wording
Friction before distracting apps	Weak to moderate (general population; uncontrolled field data plus one experiment)	one sec: opening attempts drop, plateau ~20 days (CHI 2024)	5 to 10-second pause screen during planned focus blocks, showing the current task
Temptation bundling	Moderate (general)	Gym visits +51% (Milkman 2014); taught at scale +10 to 14%, 17 weeks (Kirgios 2020)	Pair a dull task with a treat that only unlocks during it
Forgiving progress (no hard streaks)	Moderate for the harm of broken streaks; weak for repair features	Broken streak: 65.8% → 48.0% continue; repair option restores 85.2% (Silverman & Barasch); one missed day doesn't disrupt habit formation (Lally)	"X of the last 7 days", grace days, one-tap fresh start

Technique	Evidence	Key numbers	How the app uses it
Self-forgiveness after a miss	Moderate (students)	Forgiving yourself for procrastinating cut later procrastination, n = 134 (Wohl 2010)	Neutral re-plan flow after every missed block
DBT-based skills	Moderate	Executive function ES 0.64, symptoms 1.01, no gain on emotion regulation vs usual care (BMC 2022)	Short "what got in the way?" prompts; no emotion-regulation claims
Gamification	Small effect, decays	+489 steps a day, "trivial" (36 RCTs)	Off by default; light, rotating celebrations only
Coaching / accountability	Weak	19 studies, 2 RCTs, mostly n ⩽ 10 (CHADD); human support beats automated reminders (Mohr 2011)	Optional share with a buddy, coach or clinician
Body doubling	Popular, unproven	Surveys and theses only (TACCESS)	Optional "working alongside" presence; no efficacy claims
Pomodoro, "just 5 minutes", dopamine menu	Popular, unproven for ADHD	Fixed breaks cut fatigue but not learning (Biwer 2023); in a 2026 study self-set breaks gave more energy and less distraction, and were preferred (2026)	Configurable timer presets, never mandated

4.2 Design principles that follow

1. **Point of performance.** The cue arrives where the task happens: Watch haptic, Lock Screen, Dynamic Island. Barkley: reject interventions "that do not involve helping patients with an active intervention at the point of performance" (fact sheet).
2. **Micro-sessions.** Inflow users opened the app ~3.4 times a week for ~3.4 minutes and 52.7% remained at week 7 (PLOS 2022). Design every interaction for under 3 minutes.

3. **Immediate, real rewards.** Small concrete feedback at completion beats distant badges.
4. **Sparse, high-value interruptions.** Real-time alerts only for leave-now, transition and medication moments; everything else batched.
5. **Nothing punishes a miss.** No red counts, no broken-streak messages, one-tap amnesty for overdue items.
6. **Minimum viable tracking.** Burden is the top reason people abandon self-tracking (Epstein 2016). Default to passive data plus one-tap check-ins; tell users when a routine no longer needs reminders.
7. **Cognitive accessibility.** Follow W3C COGA: plain words, one primary action per screen, nothing that relies on memory, no timeouts, deep personalisation (W3C).
8. **Co-design.** HCI reviews find ADHD tech is deficit-framed and rarely built with ADHD users (Spiel 2022). Run a paid ADHD beta panel from the first prototype.
9. **Measure function, not engagement.** Akili won FDA clearance on an attention-test endpoint, then sold for about US$34 million after a ~US$1 billion listing (MedTech Dive). Track planned-vs-done, on-time rate and weekly self-rated functioning.

5. New features beyond the original inventory

The original inventory is a strong scheduling engine for one disciplined user; the research adds about 45 features aimed at the moments where ADHD actually breaks a plan. Most are small. The ones that matter most are one-tap dose logging with a double-dose guard, the "ideal vs baseline" day, lapse-proof re-entry, waiting-mode handling and transition haptics. Phase key: **MVP** = first TestFlight; **v1** = App Store launch; **v1.x** = within 6 months; **Later** = after product-market fit.

5.1 Medication and body (detail in sections 6 to 8)

Feature	What it does	Struggle it answers	Phase
One-tap dose log with double-dose guard	Log from Watch, Control Center, Action button, widget, Siri or an NFC sticker on the bottle. A second log within the same day shows "Already logged 08:02, 3 h ago" and asks for a deliberate confirm	Forgetting doses; "did I already take it?"	MVP

Feature	What it does	Struggle it answers	Phase
Dose glance	"Logged 08:02 · 3 h ago · usual focus to ~5:40 pm" on the Lock Screen Live Activity and Watch complication	Same, plus time blindness	MVP
Wake-anchored dose prompt	The prompt follows actual wake, not 8:00. Wake is detected from the app's own AlarmKit wake alarm being dismissed, a Shortcuts "Wake Up" automation, or the first Watch interaction. Apple exposes no API to read the Sleep Schedule	Variable wake time, delayed sleep phase	MVP
Personal cut-off display	User or prescriber enters a latest-dose time. Logging after it shows a neutral note: "Later than the cut-off you set. Your prescriber's instructions apply." No advice	Late dose reaching bedtime	MVP
Planned breaks	Pre-planned medication-free days are logged as planned, never "missed", and don't pollute learning	Drug holidays counted as failures	MVP
Meals and water on the medication timeline	Breakfast prompt in the morning routine, a scheduled lunch "even if not hungry", a snack before the usual end of the focus window, dinner prep. One-tap "ate something?"	Appetite suppression, dry mouth, weak hunger sensing	MVP
Discreet mode	Neutral notification text ("Morning ✓"), no drug names on Lock Screen or in calendar titles, alternate icon, Face ID lock	Stigma; calendars are never end-to-end encrypted	MVP

Feature	What it does	Struggle it answers	Phase
End-of-focus "landing" alert	30 min before the end of this user's usual focus window (learned from their own check-ins): wrap up, snack, move hard conversations. Never mentions the medication or suggests a dose	Rebound irritability	v1
Supply, repeats and script tracker	Counts down on each log, forecasts run-out, tracks repeats and script expiry, stores eScript tokens, books the prescriber visit before the last repeat	Running out; refill admin	v1
Shortage mode	Pharmacy call list with stock notes and call-back reminders, link to the TGA shortage database, prompt to contact the prescriber	Shortages (2023 to 2025 in Australia)	v1
Shareable log export	One-page PDF the user can choose to show their doctor: their own logged times, check-ins by hour, appetite, sleep and notes. No interpretation, no effectiveness claims	Short appointments; hard to recall how the weeks went	v1
Cycle overlay	Optional HealthKit cycle import; focus and mood by phase, framed as "worth raising with your prescriber"	Phase-linked changes in effect	v1.x
Caffeine log	Optional, with a personal cut-off and a sleep comparison	Caffeine use-disorder risk ($\beta = 0.35$)	Later

5.2 Time and transitions

Feature	What it does	Struggle it answers	Phase
Now / Next with a shrinking ring	One current block, one next block, time left as a depleting ring in the Dynamic Island and on the Watch	Time blindness	MVP
Transition haptics	10, 5 and 1-minute taps before a switch, plus a one-tap "park this thought" capture	Task switching (d = 0.94)	MVP
Waiting-mode protocol	Before any appointment: "You're free until 1:40. Leave-by alarm is set." Offers 2 to 3 side quests sized to the gap	Waiting mode	MVP
Leave-by with learned lateness	Works back from arrival time using travel ETA plus this user's own measured lateness buffer	Lateness	v1
Hyperfocus guardrail	During focus blocks, taps with elapsed time; after a set limit escalates to "stand, drink, eat"; warns when the next fixed item is at risk	Hyperfocus	v1

5.3 Starting and finishing

Feature	What it does	Struggle it answers	Phase
First-step breakdown	On-device model splits a task into steps and forces a first physical action under 10 minutes	Initiation paralysis	MVP
Start-tiny timer	A 2 to 5-minute timer on the first step only	Procrastination	MVP
Distraction parking lot	One tap on Watch or Lock Screen captures the stray thought and returns to the task (Safren module 2)	Distractibility	MVP

Feature	What it does	Struggle it answers	Phase
If-then plans	"When I [cue], I will [first action]", with the cue wired to a real trigger (arrive home, lunch logged, Focus on)	Prospective memory	v1
Movement primer	Suggests 10 to 20 minutes of movement right before deep work; Watch workout confirms it	Focus (acute exercise g = 0.55)	v1
Reward pairing	User-built reward list; pair a dull task with a treat that unlocks only during or after it	Delay aversion	Later
Working alongside	Opt-in, low-pressure presence ("4 people are starting a study block now") or a Focusmate hand-off	Initiation, isolation	Later

5.4 Emotion, lapses and coming back

Feature	What it does	Struggle it answers	Phase
Brain-weather check-in	One tap: low, okay, good. Low collapses the day to its baseline version and halves notifications	Emotional dysregulation, overwhelm	MVP
Ideal vs baseline day	Every day is planned twice: the full plan and a minimum viable plan (the 1 to 3 things that keep the week on track). Hitting baseline counts as a win	All-or-nothing thinking	MVP
Overdue amnesty	One tap reschedules or archives everything overdue; no red counts anywhere	Doom piles, shame	MVP
Welcome-back flow	After 3+ days away: archive stale items, pick 3 things, back in under 30 seconds, no mention of missed days	Novelty-driven abandonment	MVP

Feature	What it does	Struggle it answers	Phase
Forgiving progress	"5 of the last 7 days", automatic grace days, one-tap fresh start; no daily streaks	Streak-break abandonment	MVP
Neutral re-plan after a miss	New slot offered straight away, optional 20-second reframe	Self-blame (self-forgiveness evidence)	MVP
Safety signposting	Persistently low check-ins surface Lifeline (13 11 14) and a "talk to your GP" prompt. Signposting only, never therapy	Depression comorbidity; distress in supply gaps	MVP

5.5 Sleep and routines

Feature	What it does	Struggle it answers	Phase
Wind-down worked back from wake	Wind-down time derived from tomorrow's required wake; a protected guilt-free leisure block earlier in the evening	Revenge bedtime procrastination	v1
Morning launch on Watch	Timed sub-steps: dose, water, breakfast, hygiene, leave-by, with haptic progression	Morning start-up difficulty	v1
Routine graduation	When a routine has run unprompted for weeks, offers to drop its reminders	Tracking burden	v1.x

5.6 Life admin

Feature	What it does	Struggle it answers	Phase
Leaving-home check	Geofence exit from home: "Keys, wallet, meds?"	Losing items; forgotten doses away from home	v1
Admin sprints	10 to 15-minute timed sprints with a templated breakdown for paperwork	Doom piles, admin	v1

Feature	What it does	Struggle it answers	Phase
Health admin recall	GP script visit before the last repeat, dentist and GP recall intervals	Missed appointments	v1
Australian admin pack	PBS Safety Net tracking (resets 1 January), $25 general / $7.70 concession co-payment, state prescribing notes	Cost, cross-state rules	v1
Bills and trials on the timeline	Due dates and free-trial expiry alerts before the charge	ADHD tax	Later
Impulse cool-off	Share a product into the app; it asks again in 48 to 72 h	Impulse spending	Later
Reply-later queue and birthdays	Messages saved from any app via the share sheet; birthdays from Contacts with a 7-day gift lead time	Social upkeep	Later

5.7 Attention protection and presentation

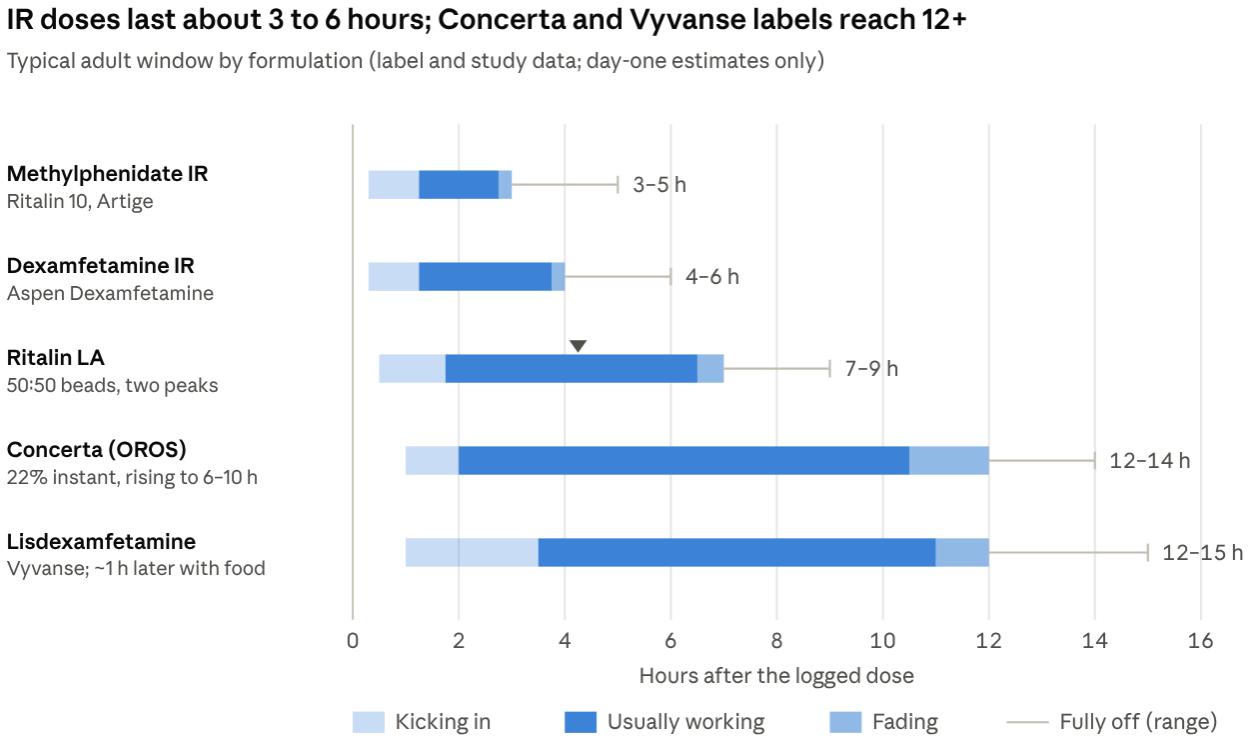
Feature	What it does	Struggle it answers	Phase
Notification budget	Default 5 interruptions a day; everything else in 2 to 3 digests; only leave-now, transition and medication alerts break through	Notification fatigue	MVP
Calm and Spark modes	Calm (default): predictable layout, muted colour, no confetti. Spark: rotating themes and celebrations	Anxiety/autism vs novelty seeking	Calm in MVP; Spark in v1
Glanceable-first surfaces	Widgets, complications, Controls and Live Activities carry the day; the app is for planning	"Ceases to exist once closed"	MVP

Feature	What it does	Struggle it answers	Phase
Setup presets	Minimal, Gentle nudges, High structure; working defaults with zero configuration	Setup burden	MVP
Focus filter link	Entering a Study Focus starts the planned block and quiets nudges	Distraction	v1
Micro-psychoeducation	60-second clinician-reviewed cards in context (sleep, appetite, routines, preparing for appointments)	Late diagnosis, poor psychoeducation	v1
Simple mode	An Assistive Access scene with three buttons: Now, Next, Log	Overload days	v1.x
Focus shields	Screen Time shields on chosen apps during planned focus blocks, with a 5 to 10-second pause screen showing the current task	Doomscrolling (OR 5.25)	v1.x (needs Apple entitlement)

6. Medication engine, part 1: what the medicines actually do

Onset, length and even the shape of the effect differ by formulation, and real users report much shorter windows than labels: 57% of 1,327 lisdexamfetamine users said theirs lasted 7 hours or less, against a label showing effect to 14 hours (ADxS survey, secondary). So label figures are only day-one estimates. Within a week or two of check-ins the user's own data should replace them.

6.1 Day-one starting estimates



FDA labels, Medsafe Concerta data sheet, AU consumer medicine information, Wigal 2010 · 5 formulations

The triangle on Ritalin LA marks a possible dip at about 4 to 4.5 hours, between its two bead releases. Lisdexamfetamine's adult effect was significant at every time point from 2 to 14 hours, but 14 hours was simply the last point tested (Wigal 2010). Full wear-off ranges deliberately run past the labelled durations (Ritalin LA 6 to 8 h, Concerta 12 h), because lab studies stop measuring at their last time point. This chart is for the build team: users never see onset or wear-off language (section 8). Concerta shows an early small peak at 1 to 2 hours, then rises to its maximum at 6 to 10 hours (FDA label).

6.2 Formulation catalogue for Australia, 2026

Medicine	AU brands	How it releases	Shape over the day	Food effect	PBS for adults (Feb 2026)
Methylphenidate IR	Ritalin 10, Artige	Immediate tablet, 2 to 3 doses a day	One fast hump	High-fat meal raises exposure ~25%	Yes, no age limit

Medicine	AU brands	How it releases	Shape over the day	Food effect	PBS for adults (Feb 2026)
Methylphenidate LA	Ritalin LA	50% instant beads, 50% delayed beads	Two peaks ~4 h apart, possible dip between	High-fat meal: longer lag, second peak ~25% lower; 40% alcohol dumps 98% of the dose in the first hour	Yes, after IR response shown
Methylphenidate OROS	Concerta, Methylphenidate-Teva XR	Osmotic pump with 22% instant coat	Small early peak, then rising to a late peak	None	Only ages 6 to 18 diagnosed in childhood
Dexamfetamine IR	Aspen Dexamfetamine 5 mg	Immediate tablet, 2 to 3 doses a day	One hump; blood levels outlast the felt effect	Vitamin C and fruit juice lower levels; antacids raise them	Yes, no age limit
Lisdexamfetamine	Vyvanse capsules 20 to 70 mg	Prodrug converted in the blood	Smooth, broad plateau	Peak ~1 h later with food	Yes, including adult diagnosis
Atomoxetine	Generic atomoxetine	Daily, builds over weeks	No daily window	High-fat meal delays peak ~3 h	Ages 6 to 18 only
Guanfacine XR, clonidine	Guanfacine; Catapres (off-label)	Daily, builds over weeks	No daily window; possible drowsiness	Guanfacine: avoid high-fat meals	Check current listing
Not in Australia	Adderall, Focalin XR, Qelbree, Jornay PM, Simtriyo	Various	Catalogue slot for overseas users	Various	n/a

Sources: Australian Prescriber, Feb 2026; FDA Ritalin LA label; FDA Vyvanse label; AU dexamfetamine CMI; ADHD Mums on unavailable medicines. Centanafadine (Simtriyo) was FDA-approved on 24 July 2026 but is not registered in Australia (Psychopharmacology Institute).

6.3 What shifts the window for one person

Modifier	What the evidence says	Applies to	How the app uses it
Food at dose time	Lisdexamfetamine +~1 h to peak; Adderall XR +2.5 h; bead products variable; Concerta none	Formulation-specific	One-tap "ate before?" on the dose log shifts the day's expected onset
Acidic drinks, vitamin C, antacids	Amphetamine exposure varies ~3.7-fold between acidic and alkaline urine (PBPK study)	Amphetamines	Optional covariate; effect size at dose time unquantified, so learned, not assumed
Menstrual cycle	Amphetamine felt stronger in the follicular phase, tracking oestradiol (r up to +.78) (White 2002)	Mainly amphetamines	Optional HealthKit cycle phase as a covariate; the app shows patterns, never adjusts doses
Acute tolerance	Felt effect fades before blood levels fall; a second closely spaced methylphenidate dose gave a smaller gain (Swanson 1999) (Frontiers 2017)	All stimulants	Half-life is a poor guide to wear-off; boosters get their own learned curve
Sleep debt	Stimulants can mask the brain signature of sleep loss (WashU / Cell 2025)	All stimulants	Short sleep lowers the day's planned load even when check-ins say "working"
CYP2D6 genetics	Poor metabolisers (~7% of Caucasians) get ~10× atomoxetine exposure	Atomoxetine	Not modelled (no daily window)
Brand or generic switch	FDA downgraded two US Concerta generics in 2014 for slower release at 7 to 12 h (FDA)	Especially OROS	Each brand is a separate product; a switch reopens learning for 1 to 2 weeks
Dose change	Exposure scales with dose (lisdexamfetamine linear 50 to 250 mg)	All	A logged dose change reopens learning

Modifier	What the evidence says	Applies to	How the app uses it
Caffeine, time of day, chronic tolerance	No usable quantitative data found	All	Optional logs only; the app learns whether they matter for this person

6.4 Boosters

A booster is a smaller immediate-release dose, taken later in the day to extend cover after a long-acting morning dose, and only when a prescriber has authorised it. NICE makes no recommendation on combining ADHD medicines because the evidence is thin (BSW guidance).

- **Its own curve.** Methylphenidate IR starts in 20 to 60 minutes and lasts 2 to 4 hours; dexamfetamine IR lasts 3 to 6 hours. Because of acute tolerance, a booster may feel weaker than the same tablet taken first thing, so the app learns booster curves separately from morning ones (an inference from acute-tolerance studies; boosters themselves haven't been measured).
- **External limits already exist.** Methylphenidate IR: last dose before 6 pm (FDA label). Australian dexamfetamine leaflet: a missed dose no later than 6 hours before bedtime. Vyvanse leaflet: avoid afternoon or evening (AU CMI). The app stores these per product as fixed label text in an "About this medicine" card the user opens. Plans and notifications never use them; the only automatic note is the neutral "ask your prescriber or pharmacist" line (7.7). It never generates timing.
- **What the app does.** The user records the booster as part of their prescribed plan (product, usual time). When they log it, the scheduler adds its curve on top of the fading morning curve and extends the window. It never prompts a booster, suggests one, or shortens a task list to make one seem necessary.

6.5 Wear-off is not rebound

Wear-off is the effect fading back to baseline. Rebound is symptoms swinging worse than baseline, usually irritability for about an hour (Understood). In a controlled children's lisdexamfetamine trial, rebound was actually rarer on the drug (3.4%) than on placebo (9.7%). So rebound is an optional, user-enabled check ("irritable or flat right now?") at the end of the window, not a default part of the model. No adult prevalence data were found.

6.6 Passive signals from Apple Watch

Starting medication raises average heart rate by about 5.4 bpm (Translational Psychiatry 2025), and Apple Watch heart rate error was about 2% in a validation study (Shcherbina 2017). But after lisdexamfetamine or dexamfetamine, heart rate stays raised for up to 24 hours while the felt effect fades. Heart rate is therefore weak evidence of onset or of a dose actually taken, and no use at all for wear-off. The app never alerts on heart rate in connection with medication.

6.7 Non-stimulants

Atomoxetine, guanfacine and clonidine build over weeks and give no daily window to plan around. For these users the app logs doses, keeps adherence and supply features, and plans the day from routine and energy check-ins instead.

6.8 Australian context that changes behaviour

- **NSW prescribing.** Specialists hold a class authority from 13 November 2023; trained GPs can continue stable prescriptions from 1 September 2025; the first GPs trained to diagnose and start treatment finished training in March 2026, with funded assessment services from early 2027 (sources differ on the exact dates) (ACI NSW). By November 2025 there were 560 GP authorities and 2,953 patients.
- **SafeScript NSW** monitors psychostimulants and is mandatory for the new GP prescribers (eHealth NSW).
- **Shortages.** Concerta, Teva XR, Ritalin IR and Artige were short in 2023 to 2025. The TGA judged the methylphenidate shortage "largely resolved" in June 2026, though specific strengths remain patchy and one tracker still lists a Concerta shortage notice to the end of 2026, and most emergency-import approvals lapsed on 31 January 2026 (ADHD Mums tracker). Shortage-driven switches are exactly the case where the learned curve must reset.

7. Medication engine, part 2: learning the personal window and filling it

The app learns each person's focus window from 3 to 5 one-tap check-ins a day for about a week, then tapers to one every 2 to 3 days. It places the work they choose inside that window. Every medication time is a user input; the learned window only ever moves tasks, never doses.

7.1 What the user tells the app at setup

Input	Example	How it's used
Each medicine, exact product and brand	Vyvanse 50 mg capsule; Ritalin 10 as booster	Picks the day-one curve from section 6; a brand change later counts as a new product
Usual dose time	"On waking" or 7:45	Reminder time the user sets; used only as a default until logs arrive
Their own experience	"Kicks in after about 45 min, good for about 7 hours"	Becomes the starting estimate, weighted above label data (a stated prior worth ~3 days of check-ins)
Prescribed booster, if any	Ritalin 10, usually early afternoon	Stored as part of their plan; the app schedules around it only once it's logged
Latest-dose cut-off	2:00 pm, set by the user or their prescriber	Shown as a neutral note if a dose is logged later; never used to prompt
Planned medication-free days	Weekends, or "sometimes"	Those days plan from routine and energy, and are never counted as missed
Desired productive activities	Essay drafting, case reading, tutorial prep	Tagged as deep work and reserved inside the window first

The user can edit their window directly at any time ("it lasts about 6 hours for me"). A manual edit is a hard override until they release it.

7.2 The "I took my medication" event

1. **Log.** One tap from Watch, Control Center, Action button, widget, Siri or the NFC tag. Backdating is built in: "took it at 7:40" or "about 20 minutes ago". Optional one-tap "ate first?".
2. **Guard.** If the same medicine was logged within the user's minimum interval, show the earlier log with its time and require a deliberate second confirm. Snooze or dismiss never counts as taken.

3. **Import check.** If the user also logs in Apple Health's Medications, those dose events are read (Health is read-only for third parties) and de-duplicated against the app's own log by time and product (WWDC25 session 321).
4. **Show.** Start the Live Activity: "Logged 8:10 · usual focus 9:10 to 5:10". It mirrors to the Watch Smart Stack and the Mac menu bar. A long-acting dose outlives a single Live Activity (8 hours active, 12 visible), so a second one is chained near the 8-hour mark.
5. **Replan.** Run the original inventory's section 3 flow, with the learned window replacing the fixed T to T+9h.
6. **Explain.** "Essay moved to 9:15. Lunch 1:10. Gym 5:30, after your focus window. Nothing conflicts."

7.3 The model

The default is an explainable template curve per product: effect rises from 0 at onset to full over a short ramp, holds, then falls to 0 at offset. Only onset and offset are learned per person; ramp lengths are fixed per formulation. Each check-in is a noisy yes, partly or no.

```
P(\text{yes at time } t) = \varepsilon_0 + (1 - \varepsilon_0 -
\varepsilon_1)\,E(t - t_{\text{dose}})
```

The lapse rates ε0 and ε1 (about 0.1 each) absorb the days someone taps "yes" while flat or "no" while fine. The posterior is computed exactly on a grid: onset 0 to 180 minutes and offset 2 to 16 hours, in 15-minute steps, about 740 cells. That runs in milliseconds on the phone and can always be explained in plain numbers.

Boosters are handled by adding curves, each with its own learned parameters:

```
E_{\text{total}}(t) = \min\Big(1,\ \sum_k E_k(t - t_k)\Big)
```

A pharmacokinetic (Bateman) curve is used only as a shape prior, because its absorption parameters can't be pinned down from sparse yes/no data (PMC10547630). Heart rate enters only as a low-weight extra signal for onset. It never triggers a prompt and never sets the end of the window.

7.4 Check-ins that people will actually answer

- **Learning phase, first 5 to 7 medicated days:** 3 to 5 single-question prompts a day on the Watch or Lock Screen: "Working now? Yes / Partly / No". Each is timed where the model is least sure, usually around the onset and wear-off edges.
- **Taper** to about one every 2 to 3 days once the 80% interval on both edges is within ±30 minutes.

- **Keep it to one question.** Questionnaire length, not prompt frequency, drives burden and data loss (163 students, 3/6/9 prompts a day, Eisele 2022). Average compliance across 477 studies is 79% (Wrzus & Neubauer 2023), so plan for 1 in 5 missing.
- **Reopen learning** on a dose, product or brand change, or when change detection fires (section 9).
- **Optional objective probe.** A 30 to 60-second arithmetic or reaction task, borrowed from the lab measure with the least placebo response (PERMP), for users who want better data. Off by default.

7.5 Showing it honestly

Evidence	What the user sees
Under 3 check-in days	"Starting estimate from what you told us: focus about 0:45 to 7:00 after logging"
3 to 7 days	"Early estimate (5 days): 0:50 to 6:20, ±50 min"
8+ days, interval under ±30 min	"Your usual focus: 0:55 to 6:30 after logging (23 check-ins, ±25 min)"
Stated and learned disagree	"You said about 7 hours; your check-ins suggest about 5.5. Use 5.5 for planning?" Accept, keep mine, or ask later

User-facing wording always says "your focus pattern from your check-ins", never "when your medication wears off" (section 8).

7.6 Filling the window with the work they chose

Every flexible task carries a focus demand: deep, medium, light or body. The window is split into zones and the solver fills them by demand.

Zone	Default use	Why
Before the dose and kicking in	Shower, breakfast, commute, a warm-up micro-task	A safe on-ramp that also defuses waiting mode
Peak (highest probability of "working")	The user's chosen deep work, longest blocks first	Largest planning deficit is in demanding work
Fading	Medium, then light tasks: admin, email, planning tomorrow	Declining effect still supports routine work

Zone	Default use	Why
Outside the window	Gym, errands, social plans, dinner, evening routines (lunch follows the user's own meal rule and often falls inside the window)	Body and social tasks don't need cover

The predicted effect E(t) also feeds the completion model as one feature among several (section 9), so the solver learns whether this person really does finish reading better at 10 am than at 3 pm. If the window can't hold the chosen work, the app changes tasks (split, defer, shrink to the baseline day, flag the deadline risk), never medication.

Worked example (a user whose learned window is 1:00 to 9:00 after logging, with a personal rule of lunch no earlier than 5 hours after the dose):

Time	Zone	Scheduled	Reason shown
7:30	Before	App-owned wake alarm dismissed; dose reminder	Reminder time the user set, anchored to wake
8:10	Logged	Live Activity starts; day replans	Dose event
8:10–9:10	Kicking in	Shower, breakfast, commute	On-ramp
9:15–12:15	Peak	Essay drafting, 2 × 80 min with a break	Deep work, longest free block
13:10–13:50	Peak	Lunch	Personal rule: at least 5 h after dose
13:50–15:20	Peak	Case reading, 90 min	Next deep task
15:20–17:10	Fading	Tutorial admin, email, tomorrow's plan	Medium and light work
16:40	—	Landing alert: wrap up, snack	30 min before usual end
17:30+	Outside	Gym, dinner, evening routine	Body and social tasks

7.7 Late, missed and unmedicated days

- **Late log.** The window simply shifts. If its projected end passes the user's own cut-off, the app shows the stored product note ("Your prescriber or pharmacist can advise how late is appropriate") and nothing more.

- **No log.** One reminder at the time the user set. If still nothing, the app offers "Plan today without medication?", which switches to a lower-demand baseline day. It never nudges the user to take a dose.
- **Planned break days** use the same unmedicated plan and are excluded from adherence and from learning.

7.8 Travel and time zones

On a time-zone change the app asks whether reminders should follow home time or local time, and suggests confirming a travel plan with the prescriber before the trip. It never picks a dose time for the user.

8. Medication engine, part 3: safety, regulation and the red lines

One feature would turn this app into a regulated medical device in Australia, the US, the UK and the EU: any suggestion about when or how much to take, including boosters. Logging doses, learning a self-reported focus pattern and arranging tasks around it can stay a productivity tool, as long as the wording, data handling and AI outputs hold that line. This is a research summary, not legal advice; the items marked **[confirm]** need an Australian health-regulatory and privacy lawyer before launch.

8.1 Where each regulator draws the line

Regulator	Fine	Crosses into device territory
TGA (Australia)	Health and lifestyle apps; self-management software that gives no "specific treatment or treatment suggestions" (HSF on the 2021 reforms)	Anything that informs a treatment decision. Diary framing doesn't help: "baby movement" diary apps are devices because of their real purpose (TGA guidance)
FDA (US)	Medication tracking with "user-configured reminders" (enforcement discretion, FDA)	Patient-facing decision support can't qualify as non-device software; January 2026 wellness guidance bars disease names and treatment recommendations in alerts (Faegre Drinker)

Regulator	Fine	Crosses into device territory
MHRA (UK)	Apps "intended to remind users that medicines are taken"; apps that replace a written diary	Software "intended to influence the actual treatment – dose … time of treatment"; indicative words include "Predicts" and "Pharmacokinetic" (MHRA)
EU (MDCG 2019-11 rev.1, June 2025)	Wellness and fitness apps	Dose recommendations to a patient are regulated software, typically Class IIa (MDCG). AI Act transparency (tell users they're talking to AI) applies either way

Two Australian cautions. The TGA's digital mental health exclusion is under review: 62% of consultation respondents called it "no longer appropriate" and removal is on the table (TGA, July 2026), so the app must not rely on it. And whether ADHD counts as a "serious" condition for the wellness exclusion is unresolved **[confirm]**, so the safe basis is that the app has no treatment purpose at all.

8.2 Hard product rules

#	Rule
1	Never suggest, calculate or prompt taking, moving, adding, skipping or changing a dose or booster, including "take your booster now to cover your 3 pm lecture"
2	Every medication reminder is set by the user at a time they choose; any schedule the app infers must be confirmed by the user
3	Dose time is a fixed input to the scheduler. When capacity is short, change tasks, never medication
4	Call the learned window "your focus pattern from your check-ins", never onset, wear-off, effectiveness or a pharmacokinetic curve
5	Never claim to treat, manage or reduce ADHD symptoms in the App Store listing, marketing, onboarding or notifications

#	Rule
6	Never alert on heart rate or sleep in connection with medication ("heart rate high for your dose")
7	Before a second same-medicine log within the user's interval, show the earlier log and require a deliberate confirm, in neutral words (never "safe to take")
8	Instant undo, plus edit and delete of any log with a local audit trail
9	Never mark a dose taken from a notification without an explicit "taken" tap; snooze and dismiss are not taken
10	Never create booster, catch-up or late-afternoon medication prompts; a late user-set reminder gets at most the stored "ask your prescriber or pharmacist" note
11	No logging for someone else, no sharing or gifting features, no marketing to undiagnosed people
12	AI outputs structurally cannot contain medication advice (8.4)
13	Medication, check-in and Health data are processed on the device or through Apple's Private Cloud Compute, never sent to third-party cloud AI
14	Show "not medical advice; follow your prescription and check with your doctor or pharmacist" in onboarding and on medication screens (App Store 1.4.1)
15	A static, non-AI "Get support" screen and deterministic crisis routing: Lifeline 13 11 14 and 000 in Australia, 988 in the US (Lifeline)
16	Launch 18+ only (8.6)

The full set of 32 rules with a source for each is in the regulation research notes saved to this Project; these 16 are the ones that shape the product.

8.3 Wording that stays on the right side

Say	Don't say	Why
"Log your doses and see your own history"	"Tracks how well your medication is working"	Diary vs monitoring treatment effect
"Reminders at the times you choose"	"Smart dose reminders" / "We'll tell you when your next dose is due"	User-configured vs app-determined
"Plan your day around when you usually feel most focused"	"Optimises your medication" / "Find your optimal dose timing"	Influences time of treatment
"Your focus pattern, from what you've logged"	"Predicts when your dose wears off"	"Predicts" and "pharmacokinetic" are MHRA device words
"Schedules your hardest tasks into your best focus hours"	"Know exactly when to take your booster"	Specific treatment suggestion
"A planner designed with ADHD brains in mind"	"Reduces ADHD symptoms" / "Clinically proven"	Disease and treatment claim
"Did I already log it? Check at a glance"	"Prevents double-dosing" / "Tells you if it's safe"	Safety decision claim
"Export your log to show your doctor if you choose"	"Helps your psychiatrist adjust your dose"	Clinical-use intent
"AI turns screenshots and messages into tasks"	"Ask our AI about your meds"	An LLM with a medical purpose makes the developer the device manufacturer
"Health details are processed on your device where possible"	"Your data never leaves your phone" (if any cloud call exists)	Deceptive privacy promise (FTC v GoodRx)

Basis: the MHRA, TGA and FDA guidance in 8.1, plus FTC v GoodRx for privacy promises.

8.4 AI guardrails

- **No field for it.** Every AI output is generated into a fixed structure (Apple's `@Generable` types or JSON schemas) that has no field for a medication action. A deterministic filter blocks dose, timing and booster language, and any medication question returns a fixed

referral: "That's one for your prescriber or pharmacist."

- **Provider terms force this anyway.** Google's Gemini API terms ban medical advice and any app "likely to be accessed by individuals under the age of 18" (Gemini terms). OpenAI bars tailored medical advice without a licensed professional (OpenAI). Anthropic requires professional review for high-risk healthcare uses (Anthropic AUP).
- **Untrusted inputs.** Screenshots, emails and web pages can carry hidden instructions (OWASP LLM01). Parsed content is data only: it fills fixed fields (title, time, place), every calendar write from parsed content needs a confirm, and parsed content can never create or change a medication entry or reminder.

8.5 Misuse and diversion

In a survey of 1,136 Australian university students, 4.4% had used prescription stimulants non-medically for study in the past year (Lucke 2018). The app should never make that easier. Done Global's founder was sentenced to 72 months in July 2026 over an Adderall scheme whose design included a 30-second auto-refill flow and advertising that persuaded people without genuine ADHD that they had it (DOJ). The supply tracker stays neutral (quantity left, next script), there is no "study hack" framing, and onboarding states the app is for medication taken as prescribed. It does not ask for prescription proof, which would only collect more sensitive data.

8.6 Age

The OAIC presumes people 15 and over can consent, but Gemini's terms bar under-18 apps outright, Anthropic requires age verification and safeguards for products serving minors (Anthropic), and the Children's Online Privacy Code must be registered by 10 December 2026 (OAIC). Launch at 18+ and revisit a 16 to 17 tier after the Code is final. The under-16 social media ban doesn't apply to a health-support tool.

8.7 App Store rules that bite

- **5.1.1(ix):** apps in healthcare or needing sensitive data "should be submitted by a legal entity … and not by an individual developer". Publish through a Pty Ltd.
- **5.1.2(i),** added 13 November 2025: "clearly disclose where personal data will be shared with third parties, including with third-party AI, and obtain explicit permission before doing so."
- **5.1.3(ii):** apps "may not store personal health information in iCloud". This rules out plain CloudKit sync of dose logs **[confirm with Apple]** (section 15).
- **1.4.2:** drug dosage calculators must come from approved entities. Another reason the app never computes dose or timing.

Source: App Review Guidelines.

9. Learning system

All learning is small Bayesian arithmetic that runs on the phone. Each learned number starts from a sensible default, shifts toward the user's own data as evidence builds (their data carries half the weight after about 4 observations), shows how much evidence it rests on, and can only become a standing preference after the user approves it. No surveyed scheduler documents learning personal duration multipliers or medication windows (Motion, Reclaim and Sunsama docs describe rules, user-entered durations and planned-vs-actual logs only), so this is the core differentiator (Motion; Sunsama).

9.1 What is learned, and how

Learning problem	Model	Starts from	Memory
Task durations, per category	Log of actual ÷ estimate, Normal-Inverse-Gamma; population → user → category hierarchy	Multiplier ~1.4 to 1.6× (Buehler thesis students took 1.64× their estimate)	Very slow (trait); forgetting 0.99 a day or none
Completion by time and context	Bayesian logistic regression, updated nightly	The user's stated preferred times, weighted as 10 to 15 observations	Per lifestyle; forgetting ~0.97 a day
Focus window	Template curve on a grid (section 7)	User's own estimate, then label data	Global per product; reset on product or dose change
Routine anchors (wake, sleep, meals)	Normal-Inverse-Gamma per weekday type	Health sleep data plus stated routine	Per lifestyle; forgetting 0.97 to 0.98 (33 to 50 days)
Transition and travel buffers	Same, per place or transition ("home after work → first deep task")	15 minutes; maps ETA for travel	Per lifestyle
Notification timing	Thompson sampling with a "send nothing" option, send probability clipped to 10 to 80%, fatigue term	3 to 6 decision points a day, gated by Focus, class, driving, sleep	Per lifestyle; forgetting 0.95 (~20 days)

Personal data's share of an estimate is n ÷ (n + κ), where κ is the default's weight in pseudo-observations. With κ = 4 that's 50% at 4 observations, 71% at 10 and 86% at 25 (Murphy; the HeartSteps study used population priors this way in production, Liao 2019).

9.2 Durations: booking realistic blocks

For each category the model predicts a multiplier on the user's own estimate, as a distribution rather than a single number:

```
\text{block} = E \times \exp\!\big(m_n + t^{-1}_{2a_n}(q)\, s_n\big)
```

- **q = 0.5** (median) for normal tasks and **q = 0.8** for deadline-critical work. Flyvbjerg's road projects needed +15% at median confidence and +32% at 80% (arXiv).
- **Day buffers come from means, not medians.** In a real estimate-vs-actual dataset the median overrun was 1× but the mean 1.81×, because a few tasks blow out (Bernhardsson). The app simulates the day's total and holds the gap between the P80 total and the sum of medians as one floating buffer block.
- **Show past actuals while the user estimates.** "Your last 6 essay-drafting sessions ran 1.4 to 2.3× your guess." In Buehler's study, linking predictions to past experience removed the bias and on-time completion rose from 29.3% to 60% (Buehler 1994). Always logged actuals, never recalled ones, which run short.
- **Learn the spread, not only the bias.** ADHD timing research shows more variable estimates, not just shorter ones (Marx 2021).
- **New categories** start at the user's overall multiplier, not the population's.

9.3 The behaviour ledger, extended

The original ledger fields stay. Add these, because the learning depends on them:

New field	Why
Lifestyle and mode active	Routines and time-of-day models are learned per lifestyle (section 10)
Focus-window position (predicted effect at the time)	Lets completion be modelled against the window
Reason code for anything that didn't happen	Separates rule-driven skips from real misses
Outcome source: tap, timer, Watch workout, inferred	Weights passive evidence below explicit logs
Day logging coverage	Stops good-days-only logging from biasing the model

New field	Why
Decision propensity	Makes offline replay of experiments valid (9.7)

Reason codes (one tap, shown only when something lapses):

1. Blocked by a rule or event (clash, class ran over, exam) → excluded from learning.
2. Chose not to, or couldn't start → counts against that slot and context.
3. Started, then stopped → duration known to be at least the elapsed time (censored), never treated as done.
4. Did it but didn't log → success; asks for the real duration once.
5. No answer → missing.

Never default "actual = planned" for learning (Sunsama offers this as a display setting; it would pull every multiplier toward 1). Days where under 30% of planned items have an outcome are excluded from completion models. A gap of 3+ days away freezes forgetting, so the model doesn't decay toward defaults while the user is absent.

9.4 Confidence and approval

Evidence	Status shown	Can it change the plan?
Fewer than 3 observations	"Typical for people like you: ×1.4"	Yes, as a soft default
3 to 7	"Early estimate (n = 5)" with a wide range	Yes, as a soft cost
8+ and range narrower than ×1.5	"Learned"	Yes, as a soft cost
Proposed standing rule ("move reading to mornings")	Suggestion with its evidence	Only after approval; needs 10+ comparable events in the current lifestyle, 90% probability the effect is real and worthwhile, and 2 weeks of stability

Stated preferences are strong priors; behaviour can only propose changes to them. A rejected proposal is shelved for 30 days. A "What the app has learned" screen lists every learned value with its evidence count, range and last change, and a reset button for each.

9.5 Experiments without superstition

- **Formal questions** ("90 vs 60-minute blocks") run as a blocked N-of-1 comparison: pairs of study days, one of each in random order, for 6 to 8 pairs, analysed with a Bayesian model (AHRQ N-of-1 guide). Adopt only if the probability of a worthwhile gain exceeds 80%, and only with the user's yes.
- **Everyday exploration** is limited to flexible, low-stakes items (errands, habits, optional study), at most 1 to 2 exploratory moves a day, with posterior scaling to damp "weird suggestions" (Chapelle & Li).
- **Never explored:** exams and assessments due within 48 hours, medication reminders, fixed commitments and anything the user marks protected.

9.6 Noticing that life changed

Bayesian online change-point detection runs on wake time, sleep midpoint, first activity, focus-window reports and duration ratios (Adams & MacKay 2007), with a cheap CUSUM alarm alongside (NIST). A shift counts as real only if all four hold:

1. It's probably under 3 days old (probability over 0.8).
2. It moves the mean by more than 30 minutes or one normal day-to-day spread, whichever is larger.
3. At least two independent signals agree (say, wake time and first Watch activity).
4. It has lasted 3+ days.

Known triggers (a logged dose or product change, a new timetable in the calendar) make the model expect change for 14 days so it adapts fast without being told. Every detected change is explained: "Since Tuesday your wake time averages 9:10, up from 7:20. Update your routine?"

9.7 Proving the learning helps

- **Durations:** share of tasks finishing within the P50 (target 50%) and P80 (target 80%) predictions over a rolling 30 tasks, against the raw-estimate baseline, where only about 30% finish within their best guess.
- **Completion predictions:** Brier score and a 5-bin reliability check.
- **Plan-level:** blocks done within 30 minutes of plan, reschedules a day, end-of-day overrun minutes, suggestion acceptance.
- **Offline replay:** every exploratory decision logs its probability, so a new model version can be replayed over the user's history before it ships (Li et al.).

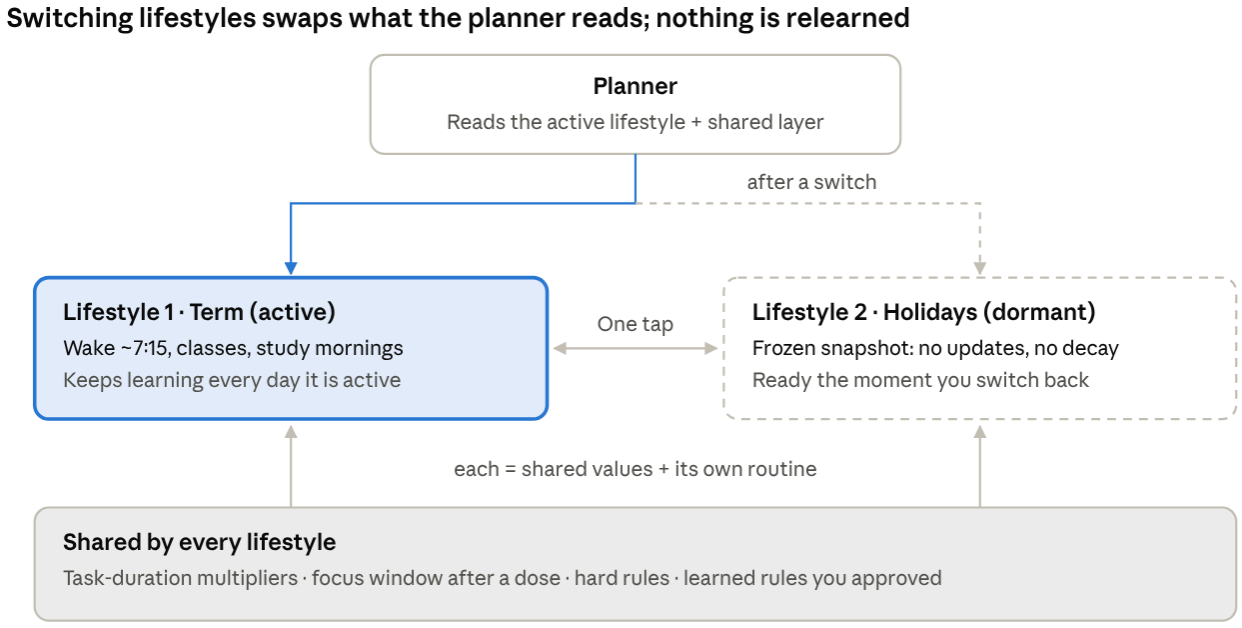
9.8 Where the language model fits

The LLM never computes a number. It turns "that was stupid" into a targeted question ("Was the problem research after gym, or the block being too short?"), phrases explanations from numbers the deterministic code produced, and extracts structure from text and screenshots. That matches the original inventory's split: the LLM interprets, the rules engine constrains, the solver solves.

10. Lifestyles: the Lifestyle 2 toggle

Lifestyle 2 is a second, separately learned version of the user's routine. Flipping the toggle changes which version the planner reads, so switching back is instant and nothing is forgotten. A brand-new Lifestyle 2 starts from the user's own weekend pattern plus three quick answers, and its routine is about 50% personalised after 3 days, 70% after 7 and 82% after 14.

10.1 How it works



Lifestyle switching · 2 lifestyles, 1 shared layer

Only the active lifestyle learns. The dormant one is frozen exactly as it was, with no updates and no decay, so returning to it is a pointer swap rather than a relearn. Old models are stored and reused because routines recur, the standard approach for "reoccurring" drift (Lu et al. 2020).

10.2 What belongs to a lifestyle, and what's shared

Kept per lifestyle	Shared across lifestyles
Wake, sleep and meal anchors, by weekday type	Task-duration multipliers (with a small per-lifestyle offset for interruptions)
Usual dose clock time (it follows wake time)	Focus window measured from the dose: pharmacology doesn't follow the calendar
Time-of-day completion curve (an offset on a shared base)	Task-category effects (reading vs admin)
Notification timing and receptivity	Hard rules, unless the user tags one to a lifestyle
Routine set, and which items are calendar-worthy	Learned rules the user already approved, unless tagged
Priority ladder order (holidays might rank shifts > gym > study)	Places, travel overheads, opening hours
Movement policy and preferred slots	Medication list, supply and scripts
Weekly template (classes vs none, shift patterns)	The behaviour ledger itself (every event is tagged with its lifestyle)

10.3 The first time Lifestyle 2 is switched on

1. **Three questions, under a minute.** Roughly when will you wake? Any fixed commitments (work shifts, a course, travel)? What do you want this period for (rest, work, a project, catch-up)? Each answer becomes a stated starting point.
2. **Borrow the user's own free-day pattern.** The starting wake and sleep anchors come from their term-time weekend timing, which is already measured. With no weekend data, the default is weekday wake plus 50 to 90 minutes. University students released from fixed schedules went to bed about 50 minutes later on weekdays (CU Boulder on Wright 2020).
3. **Weak prior, wide spread.** The borrowed pattern is worth only about 3 days of evidence, with 1.5× the usual uncertainty, so real data takes over fast.
4. **Fast forgetting for 14 days.** Holiday sleep in Melbourne adolescents shifted abruptly, kept drifting later through the first week and settled in the second (Bei 2014). The model discounts older days quickly until it settles, then slows down.
5. **Plan lightly while unsure.** Early Lifestyle 2 days lean on "anytime today" tasks and the baseline day rather than tight time blocks, and tighten as confidence grows.

Days into a new Lifestyle 2	Share of the routine estimate from the user's own data
3	~50%
7	~70%
14	~82%

10.4 Switching back

Lifestyle 1 is exactly as it was on the day of the switch. Irregular sleep delays the body clock (melatonin onset was 2 h 36 min later in irregular sleepers, Phillips 2017), so the first 3 to 5 days back carry extra uncertainty and a lighter plan. An optional "back to term" ramp moves the app's own wake alarm earlier in small steps across the last few days of the break.

10.5 Next holidays

The next switch to Lifestyle 2 restores last time's snapshot with its confidence reduced in proportion to how long ago it was (capped at about a week of evidence). Last summer informs day one, and this summer's data dominates within about a week.

10.6 Scheduling a switch in advance

The user can enter session dates once ("Lifestyle 2 from 28 Nov to 22 Feb") and the app switches on those dates, with a heads-up the evening before. The toggle still works at any time.

10.7 Catching a forgotten toggle

- **Both lifestyles learned:** a two-state filter compares each day's wake time, sleep midpoint, first activity, class hours and (if permitted) campus visits against both lifestyles. It is deliberately sticky (97% chance of staying put), so one late night never triggers it. It asks when the other lifestyle has been over 90% likely for 2 days running (Fox et al., sticky HMM).
- **Lifestyle 2 never used:** change-point detection on the routine anchors, made more sensitive around entered session dates.
- **Always a suggestion, never an automatic flip:** "Last 3 days: up at 9:40 (term average 7:15), no classes. Switch to Holidays? Your settings from July are ready." One tap to accept or dismiss.

- **Late switch:** "Switch since Monday" reassigns the logged days since Monday to Lifestyle 2 by replaying the raw events.

10.8 Lifestyles vs modes

	Lifestyle	Mode
What it is	A learned routine baseline	A temporary rule overlay
Examples	Term, Holidays, Placement, Night shifts	Exam, Sick day, Travel, Assessment crunch
Learns?	Yes, its own anchors and timing	No; it changes priorities and rules
Typical length	Weeks to months	Days to a few weeks
Combine?	One active at a time	Stacks on whichever lifestyle is active

Cap named lifestyles at about 4. If two turn out statistically near-identical, the app suggests merging them.

10.9 The toggle itself

It lives in Control Center, on a widget and at the top of the Today screen, and it can be renamed ("Term" and "Holidays" read better than "Lifestyle 1 and 2"). Tapping it shows a one-screen preview before applying: "Wake anchor 9:30 instead of 7:15 · study blocks move to afternoons · 6 term-only routines paused." Term-only routines and app-created blocks are paused, never deleted; external calendar events like classes are left alone. Undo is one tap.

10.10 Where it can go wrong

Risk	Mitigation
A one-week mid-semester break gives too little data	Under 3 days of evidence, stay close to the prior and plan loosely; don't create a new lifestyle
Mixed periods (holidays with some shifts)	Shifts are fixed events inside Lifestyle 2; the lifestyle is about the anchors, not the calendar

Risk	Mitigation
Toggling back and forth daily	That pattern is a mode or a weekly template (weekday vs weekend), and the app suggests one
Holiday data leaking into term estimates	Every ledger event carries its lifestyle, and only shared parameters learn across both
Stale snapshot after a year	Confidence decays with time since last use, so this year's data dominates within a week

11. Apple platform integration

Target iOS 26 as the minimum and treat iOS 27 as an upgrade. iOS 26 brings everything the core needs: reading Health medications, the on-device Foundation Models, AlarmKit, table OCR and user-started long tasks. iOS 27, released 14 September 2026 (MacRumors), adds Apple's Private Cloud Compute model for third-party apps, 8K on-device context and Siri app schemas. Eight platform limits reshape the original design; each has a workaround below.

11.1 Capability map

Need	Apple API	Hard limits	Decision
Dose log and Health import	App's own store; HealthKit `HKUserAnnotatedMedication`, `HKMedicationDoseEvent` (iOS 26), per-medication permission	**Third parties can't write doses** ("Medication data is read-only", Apple forums); Health may backdate, delete and re-save events	Own log is the source of truth; optional anchored import, de-duplicated by Health's IDs
Sleep, workouts, heart rate, mood, cycle	HealthKit sleep stages, `HKWorkout`, heart rate, `HKStateOfMind`, menstrual types	Store unreadable while the phone is locked; background updates hourly on iPhone, 4 an hour on Watch with a complication; denied permission looks identical to no data	Compute on unlock or in the foreground; treat missing as unknown, never zero

Need	Apple API	Hard limits	Decision
Wake time	None: **no API for the Sleep Schedule or wake alarm** (forums)	—	App-owned AlarmKit wake alarm, a Shortcuts "Wake Up" automation calling an App Intent, or first Watch interaction
Calendar	EventKit, full access	Change notifications carry no diff; `eventIdentifier` changes when an event moves calendar; recurring events share one external ID; calendars are never end-to-end encrypted (TN3153)	App UUID in each event's URL field plus a local map; a dedicated app calendar; neutral titles
Reminders	EventKit	No subtasks, flags, sections or tags through the API, even in iOS 27	Rich tasks live in the app; Reminders mirrors simple items and location alarms
Voice, Shortcuts, Action button	App Intents, App Shortcuts, `ControlWidget`, iOS 27 app schemas (Calendar, Reminders)	No health or medication schema; Siri AI needs iPhone 15 Pro or later, English only at launch, not EU or China	"Log my dose / booster" App Shortcuts that work on every device; tasks exposed as Reminders and Calendar entities

Need	Apple API	Hard limits	Decision
Live window display	Live Activities (Lock Screen, Dynamic Island, Watch Smart Stack, Mac menu bar)	8 h active + 4 h on the Lock Screen; 4 KB state; no network; buttons need an unlock (Apple)	Start from the dose-log intent; chain a second activity near 8 h; countdowns with dynamic date text
Alerts that break through	AlarmKit (iOS 26)	Overrides Silent and Focus; iPhone and iPad only, mirrors to Watch	Opt-in only: wake alarm, a user-set "hard" dose reminder, leave-now
Ordinary notifications	UserNotifications	Only the 64 soonest local notifications are kept; critical alerts need Apple approval (unlikely for this use)	Rolling 24 to 48 h schedule, refilled on every launch and event; no drug names on the Lock Screen
Focus modes	`SetFocusFilterIntent`	The app sees only its own filter's settings	A Study Focus starts the planned block and quiets nudges
Distraction shields	FamilyControls, ManagedSettings, DeviceActivity	Apple entitlement for the app and each extension; 6 MB extension memory; threshold bug on iOS 26.2; not on Mac	v1.x; stateless extensions; register only the next few blocks
Background upkeep	`BGAppRefreshTask`, silent push, `BGContinuedProcessingTask`	Refresh gets ~30 s when the system chooses; silent pushes throttled above 3 an hour; continued tasks must be user-started	Event-driven upkeep (below)

Need	Apple API	Hard limits	Decision
Geofences	`CLMonitor`	20 conditions per app	Rotate the 20 most relevant; or hand location alarms to Reminders
Travel time and opening hours	MapKit `calculateETA` with transit	**MapKit exposes no opening hours**; Google Places has hours but must be shown on a Google map and can't be cached (Google policy)	ETA from MapKit; hours from Google Places in a compliant view, or user-entered
On-device AI	Foundation Models	4,096 tokens (iOS 26), 8,192 (iOS 27, newer devices); Apple Intelligence devices only; rate-limited in the background (TN3193)	Short structured extraction and explanations; every feature has a non-AI path
Cloud AI	Private Cloud Compute model (iOS 27, 32K context); Claude and Gemini packages behind the same API	Entitlement; free under 2M downloads in the Small Business Program; per-user daily quotas (WWDC26 guide)	PCC for health-adjacent heavy parsing; third-party models only for non-health text with opt-in
Roster screenshots	Vision `RecognizeDocumentsRequest` (tables, iOS 26); Visual Intelligence	Visual Intelligence hands apps only a pixel buffer	Vision table extraction → deterministic parse → on-device clean-up → user confirm → calendar

Need	Apple API	Hard limits	Decision
Sync	SwiftData + CloudKit, encrypted fields	App Review 5.1.3(ii): no personal health information in iCloud	Health data stays local with the app's own end-to-end encrypted sync (section 15)
Mac	SwiftUI multiplatform, `MenuBarExtra`	No Health data, no Screen Time, AlarmKit only via Catalyst	Planning console plus a menu-bar "now / next / focus window"
Low-load mode	`AssistiveAccess` scene (iOS 26), Reduce Motion, Dynamic Type	—	Simple mode: Now, Next, Log

11.2 Background upkeep without background time

iOS gives no reliable periodic runtime, so the original "continuous background maintenance" becomes event-driven. The app re-plans:

- on every foreground launch;
- inside App Intents (logging a dose, marking done, "fix my day");
- in HealthKit observer callbacks (sleep ended, workout saved);
- on calendar change notifications after launch;
- in background refresh when iOS grants it;
- from at most about 3 server-sent silent pushes an hour as a backstop.

Heavy jobs (weekly re-plan, roster import) run as user-started long tasks with a system progress indicator.

11.3 What each device is for

Device	Role
iPhone	The hub: model state, planning, Live Activities, notifications, Health reads
Apple Watch	The point of performance: one-tap logs, check-ins, transition haptics, complication (which also buys 4 background wakes an hour), double-tap to log

Device	Role
Mac	Planning console: week view, unscheduled queue, forecasts, rule editor, roster import, history
iPad	Same as Mac, plus Health and Screen Time

11.4 Reach

iOS 27 runs on iPhone 11 and later, but Apple Intelligence needs iPhone 15 Pro or later. Many students carry older phones, so every AI feature must degrade to a manual or rules-based path. Whether Siri AI is live in Australian English at launch is unconfirmed.

11.5 Test in the first two weeks

- Medication read permission (one developer reports a crash that may be an entitlement gate).
- Transit ETA quality for Sydney through MapKit.
- Screen Time entitlement turnaround.
- Whether a health-adjacent app qualifies for the Private Cloud Compute entitlement.
- App Review's reading of 5.1.3(ii) for user-entered dose logs.

Also confirm that the app can reliably see its own AlarmKit wake alarm being stopped, since wake-anchored reminders depend on it.

12. The original feature inventory, reviewed

The inventory's architecture holds up: "the LLM interprets, the rules engine constrains, the solver solves, the connectors execute, the behaviour engine learns" is the right split. Its main blind spots are four. It assumes continuous background runtime that iOS doesn't give. It assumes write access Apple doesn't grant (Health doses, rich Reminders). It asks an ADHD user for a lot of configuration. And it treats the medication window as a fixed rule rather than a learned, regulated-sensitive estimate. Numbers in brackets are the inventory's section numbers.

12.1 Interaction layer [2, 17, 64–66]

Feature	How it works on Apple	Limits and failure modes	Plan
Chat and short commands [2]	On-device model parses text into fixed command types; a deterministic grammar handles the common short forms ("gym tomorrow 3", "took meds", "skip gym") with no model at all	4K-token context on iOS 26; no Apple Intelligence on older iPhones; ambiguous phrasing	Grammar first, model second, PCC for long inputs; show the interpretation before any medium or high-risk change
References ("move that", "same as last Tuesday") [2]	A deterministic resolver keeps a stack of recently mentioned items and looks up dated history	The model can pick the wrong "that"	Resolve in code; if two candidates are close, ask with both shown
Voice and Siri [2, 66]	App Shortcuts with fixed phrases work on every device, Watch and AirPods; iOS 27 app schemas let Siri AI act on tasks and events	No medication schema; Siri AI needs iPhone 15 Pro+, English only, not EU/China	Fixed phrases for all core intents; Siri AI is a bonus, never required
Screenshots and photos [2, 10]	Vision table OCR on-device, then parsing into fixed fields	Merged cells, other people's rows, handwriting, dark mode screenshots	See roster ingestion (12.3)

Feature	How it works on Apple	Limits and failure modes	Plan
Email and booking ingestion [2]	Share sheet from Mail or Gmail into the app; optional Gmail or Outlook connection later	No Apple Mail API; Gmail's restricted scopes go through Google's app verification, which may require an independent security assessment [confirm]; emails can carry hidden instructions	Launch with share sheet and on-device parsing only; every extracted booking needs a confirm; connectors after launch
Calendar and reminder edits by the user [2, 15]	Detected on the next launch or change notification, then diffed against the app's map	iOS gives no diff and may not wake the app	Full diff of the planning horizon on each wake; consequences surfaced as one summary, not a stream
Fix my day [17]	Interpret → apply rules → update estimates → solve → preview → execute → verify → explain	Model latency; big reshuffles erode trust	Deterministic intents for the 20 most common phrases; a stability penalty so only affected blocks move; one-tap undo
Widgets and Watch [2, 64]	WidgetKit and complications carry Now, Next and the focus window	40 to 70 widget reloads a day; widget buttons dead while locked	Dynamic date text for countdowns; Watch and AlarmKit actions for locked-phone logging
Daily briefing and weekly planning [2, 30, 71]	Briefing fires after detected wake, not a fixed time; weekly planning is a guided 5-minute session	Walls of text get skipped	Max 5 lines, the day's 1 to 3 priorities first; weekly session is optional and resumable

Feature	How it works on Apple	Limits and failure modes	Plan
End-of-day review [2, 72]	Asks only about ambiguous items	Nightly questions become a chore	At most 3 one-tap questions; skipped reviews are fine (reason code "no answer")

12.2 Scheduling core [3–9, 18, 30–32]

Feature	How it works on Apple	Limits and failure modes	Plan
"I took my medication" [3]	Section 7.2	Fixed T to T+9h is wrong for most people; late wake; backdating; duplicate logs; Health import duplicates	Learned personal window; backdating; double-dose guard; de-duplication; dose time never adjusted by the app
Task properties [4]	22 fields per task, stored in the app	Asking an ADHD user for 22 fields kills adoption	Ask only for title, rough duration and deadline; infer category, energy, min and max block, split and location from the title and history; everything editable later
Solver [4]	A small constraint solver on-device (for example Google OR-Tools CP-SAT compiled for iOS, or a custom search), with hard constraints, then weighted soft costs; validate the choice in a spike	Re-solving can reshuffle the whole week, which feels chaotic	Add a "minimal change" cost, freeze the next 2 hours, and move as few blocks as possible

Feature	How it works on Apple	Limits and failure modes	Plan
Conflict resolution [5]	Priority ladder plus explicit exceptions; never silently double-book	Frequent "Ask" prompts cause decision fatigue; outside changes (a new meeting invite) can double-book without the app	Batch questions into one card with a recommended answer; an unanswered question keeps both items visible, never a timeout; external clashes flagged, not auto-fixed
Gym logic [6]	Generalised movement policy: candidate slots, skip conditions, rule-based skips excluded from learning	Rule-based skipping can remove all exercise in exam weeks, when it helps focus most (acute exercise g = 0.55)	A weekly movement floor with short fallbacks (20-minute walk before study); Watch workout detection auto-credits
Meals and routines [7]	Meal anchors on the medication timeline; routines tagged calendar-worthy or checklist	Appetite suppression means lunch reminders get ignored	Lunch as a protected block; pre-window-end snack; one-tap "ate"; checklist habits live in a widget, not the calendar
Uni study system [8]	Assessments with weight, workload, dependencies and focus demand	Manual entry of every assessment is heavy	Import from the unit guide PDF (Vision + on-device model) and the LMS calendar feed where one exists; confirm each item
Deadline and exam engine [9]	Works back from the deadline in blocks sized with the user's P80 multiplier; Exam mode as an overlay	Estimates start wrong; long assessments still get left late	Early forecast ("2 hours short by Thursday") paired with a proposed fix; milestones visible on the Lock Screen widget

Feature	How it works on Apple	Limits and failure modes	Plan
Partial completion [18]	"Done for now" records time spent and remaining minutes	Remaining estimate is itself a guess	Remaining = original estimate × learned multiplier − time spent; treated as censored data for learning
Weekly planning and forecasting [30, 31]	Capacity = free hours inside predicted focus windows × completion probability	A shortfall number can trigger anxiety	Always shown with a concrete fix; wording is neutral, never "you're behind"
Scenarios and what-if [32, 63]	Clone state in memory, solve, show the diff; nothing written until approved	Big diffs are hard to read	Show only changed items and the net effect ("gym moves to Sat, PIL 1 h short")

12.3 Integrations [10–16, 34–36, 45–47]

Feature	How it works on Apple	Limits and failure modes	Plan
Roster ingestion [10]	Screenshot → Vision table OCR → find the user's row → shifts → diff against the Work calendar → confirm → write	Finding "my row" in a table of names; every employer's roster app looks different; iOS 27 Visual Intelligence may already have added the same shifts	User taps their row once and the layout is remembered per source; idempotency key = employer + date + start; existing matching events are adopted, not duplicated; removed shifts are asked about

Feature	How it works on Apple	Limits and failure modes	Plan
Travel [11]	MapKit ETA (car, walk, transit) plus a learned per-place overhead (parking, lifts, walking in)	Sydney transit ETA quality through MapKit unverified; ETA can't run inside a Live Activity	Learned overhead on top of ETA; keep the last buffer if maps fail; test Sydney transit in week 1
Opening hours [12]	Google Places details for hours, shown on a Google map with attribution	MapKit has none; Google data can't be cached beyond place IDs; public holiday hours often wrong	Fetch fresh before scheduling; user-entered hours override; "conflicting hours, not scheduled yet" message kept
Errand clustering [13]	Small route optimisation over pending errands near an existing trip, constrained by free time and hours	iOS monitors at most 20 geofences	Rotate the 20 most relevant places; or give location alarms to Apple Reminders
Calendar ↔ task identity [14]	Every app item has a UUID; the event's URL field carries it; a local map stores external IDs and occurrence dates	Event IDs change when an event changes calendar; recurring occurrences share one external ID; Exchange IDs differ by device	Match by UUID in URL first, then external ID + start date; unmatched items go to a "needs review" list, never silently deleted
Two-way sync [15]	Diff on every wake; linked task and event move together	Reminders can't hold subtasks or tags; offline edits on two devices can collide	Rich structure stays in the app; collisions on low-risk items resolve to the latest edit, high-risk ones ask; deletions keep a 30-day tombstone for undo

Feature	How it works on Apple	Limits and failure modes	Plan
Background maintenance [16]	Event-driven (11.2), with the four notification classes kept: silent fix, inform after, ask before, urgent	iOS won't run the app on a timer	Upkeep on every wake and intent; a server push backstop at most ~3 an hour
Reminder intelligence and escalation [34, 35]	Action-based reminders ("Leave at 2:20") replace "Dentist in 10 minutes"; assessments escalate 7d → 3d → 1d → morning	64 pending local notifications; escalation adds to notification load	Rolling 48-hour notification window; escalation counts against the daily budget except leave-now and medication
Location surfacing [36]	Geofence on arrival near an errand cluster, only if open and time allows	Too many pings; battery	Max 1 to 2 location prompts a day; only when the place is open and the next hour is free
Connectors [45, 46]	Apple Calendar and Reminders via EventKit; Google and Exchange calendars through iOS accounts; Gmail, Notion, Drive and LMS feeds later	Direct Google APIs need OAuth verification; LMS APIs vary	Launch on EventKit only (it already sees Google and Exchange accounts added in Settings); add connectors one at a time with a status page
Canonical objects [47]	Kept, plus `Medication`, `Product`, `DoseEvent`, `CheckIn`, `FocusWindowModel`, `Lifestyle`, `Mode`, `ReasonCode`, `ConsentRecord`	—	See section 16

12.4 Learning and memory [19–29, 50–56]

Feature	How it works on Apple	Limits and failure modes	Plan
Behaviour ledger [19]	Local event log, tagged with lifestyle, mode and reason codes	Logging burden is the top reason people abandon self-tracking	Passive capture first (timers, Watch workouts, calendar attendance); one-tap reason codes only when something lapses
Pattern learning and confidence [20–22]	Section 9	Over-reading 2 examples	Evidence thresholds, shrinkage toward defaults, approval before any rule change
Contextualised failures [23]	Reason codes (9.3)	Users won't classify every miss	Infer "blocked by rule" automatically when the planner itself skipped; ask only for real misses
Sleep and wake [24]	HealthKit sleep stages	Only if the Watch is worn at night; unreadable while locked	Fall back to first Watch activity or the app's own wake alarm; never infer cognition from sleep
Workouts [25]	HealthKit workouts matched to planned movement blocks	Gym sessions without a Watch aren't seen	One-tap "did it"; rule-based skips excluded
Energy feedback [26]	Watch Low / Normal / High, saved to the app and optionally to Health as State of Mind	Check-in fatigue	Folded into the focus check-in during the learning phase; optional afterwards

Feature	How it works on Apple	Limits and failure modes	Plan
Health-data safety [27]	Health signals adjust discretionary load only, with "Accept adjustment?"	FDA and TGA wellness rules bar condition-linked alerts	Never name a condition or medication in a health-based notification
Duration and schedule models [28, 29]	Section 9.1 to 9.2	Categories too coarse or too fine	Start with ~10 categories; the model suggests splitting one when its spread is very wide
Memory vs behaviour; contradictions; corrections [50–53]	Stated preferences are strong priors, behaviour proposes changes, corrections become a one-off, a preference or a hard rule	—	Kept as designed; provenance shown on every learned item
Metrics, weekly review, experiments [54–56]	Section 9.5 and 9.7	A weekly list of shortcomings reads as a report card	Lead with what worked; at most 3 findings; each with one proposed change

12.5 Reliability [40–44, 60–61, 73, 76]

Feature	How it works on Apple	Limits and failure modes	Plan
Explainability [40, 74]	Every change stores the rule, cost terms and evidence that produced it	Explanations generated by a model can be wrong	Numbers and reasons come from the solver's record; the model only phrases them

Feature	How it works on Apple	Limits and failure modes	Plan
Undo and audit log [41, 73]	Before/after snapshot per change set; "restore this morning's plan"	External apps may have changed since	Restore re-diffs against current state and asks about anything edited in between
Transactional execution [42]	Plan → validate → batch saves with EventKit's deferred commit → commit → re-read	EventKit has no true multi-item transaction across accounts; a sync server can reject later	Compensating rollback from the audit log; exact unresolved changes listed if rollback fails
Post-write verification [43]	Re-read every written item after commit, and again on the next wake	iCloud and Exchange sync lag	Items stay "pending" until seen twice; a mismatch triggers a repair, not a duplicate
Idempotency [44]	Every import carries a deterministic key (source + external key)	Screenshot OCR can read "1-4" as "1-14"	Keys built from normalised fields; implausible shifts (over 12 h) require a confirm
Failure behaviour and offline [60, 61]	Keep existing buffers when maps fail; never mark a task rescheduled if the write failed; local-only mode queues changes	—	Kept as designed; the deterministic grammar keeps short commands working offline
Connector health [76]	Diagnostics page with last successful read and write per source	A silent Health permission denial looks like "no data"	Show "no data seen since" rather than "connected" for Health types

12.6 AI, privacy and cost [48–49, 57–59, 62, 75, 77–78]

Feature	How it works on Apple	Limits and failure modes	Plan
AI responsibilities [48]	Kept: the model interprets, deterministic code decides	Model outputs can smuggle medication advice	No medication-action field in any output type (8.4)
Approval policy [49]	Kept, with a new top class: medication actions are never taken or suggested	—	—
Privacy categories and local-first [57, 58]	Kept, but medication, check-ins and Health data are locked to on-device or Private Cloud Compute regardless of settings	Users may want cloud AI features that need that data	Explain the limit; send derived, non-health summaries only ("a light day", not "slept 5 h, HRV low")
Permission transparency [59]	Kept; one screen per permission, asked in context, never all at onboarding	Asking for everything up front triggers refusals	Progressive permissions: calendar first, Health when the first check-in happens, location when the first errand is added
Search and queries [62]	On-device model turns a question into a structured query; code runs it over the ledger	Model may misread the question	Show the interpreted query ("Gym sessions, last 30 days") above the answer
Export [75]	JSON and CSV of everything	An export file holding medication logs is sensitive	Warn before export; exclude Health-sourced data by default; the file stays on-device until the user shares it

Feature	How it works on Apple	Limits and failure modes	Plan
Model routing and cost [77, 78]	Grammar → on-device model → Private Cloud Compute → opt-in third-party model	Third-party AI costs roughly US$0.35 to $1.05 per active user a month (Claude Haiku 4.5 estimate, competitor research notes)	On-device and PCC are free to the developer under the PCC programme limits, so most users cost near zero; third-party AI is an opt-in feature

12.7 Time and attention [67–70]

Feature	How it works on Apple	Limits and failure modes	Plan
Running late [67]	"Running 30 late" shifts downstream flexible blocks and leave-by times; fixed items untouched	Late cascades can empty the afternoon	Offer the baseline day when more than half the flexible work no longer fits
Overrun detection and timers [68, 69]	Block timers run in a Live Activity; the Watch taps on overrun	ADHD users forget to start and stop timers	Blocks auto-start at their planned time with a one-tap "not started"; the next block's start stops the last
Notification intelligence [70]	Decision-point alerts only, within the daily budget (5.7)	Budgets can hide something important	Leave-now, transition and user-set medication reminders are exempt; everything else batches

13. Why this app could fail, and what to do about it

The most likely killers are not missing features but trust and adoption failures: users drifting off after two weeks, a scheduler that over-packs the day or moves one thing badly, and a setup that asks too much of the people it is meant to help. Medication

awareness is the product's moat and also its main regulatory and reputational exposure. The table is ordered by how likely each risk is to sink the product (my judgement from the evidence).

13.1 Adoption and trust

#	Risk	Evidence	Mitigation
1	Users drift off after 2 to 3 weeks	Mental-health apps: 3.9% still active at day 15, 3.3% at day 30 (Baumel 2019); prompt effects fade to zero by day 28 (HeartSteps); boredom r = 0.40 with ADHD	Deliver value without opening the app (Live Activity, complication, calendar blocks); lapse-proof re-entry; rotate wording and optional themes; micro-sessions under 3 minutes (Inflow, built on ~3-minute sessions, kept 52.7% at 7 weeks in an uncontrolled study); track day 1, 7 and 30 cohorts from launch
2	Setup is too heavy before any value	"Lengthy initial setup" (Routinery), "a full-time job just adding data" (Motion), "more sophisticated than intuitive" (Reclaim)	Onboarding under 3 minutes: calendar permission, medication plus the user's own estimate, three priorities. Everything else inferred or preset
3	The scheduler over-packs the day	Motion: days with "no breathing room" (Saner review); Reclaim: "overwhelming seeing my calendar all booked" (G2)	Fill at most ~60% of free focus time; mandatory transitions and a floating buffer; app blocks on a separate, hideable calendar; the baseline day is always one tap away

#	Risk	Evidence	Mitigation
4	One visibly bad move kills trust	People lose confidence in an algorithm faster than in a human after the same mistake (Dietvorst 2015)	Suggest-then-confirm for the first 2 weeks; never move pinned or user-created items; a reason on every placement; one-tap undo; cap automatic moves per day; a daily "what moved" digest
5	The app causes shame	Emotional dysregulation g = 1.17; broken streaks cut continuation from 65.8% to 48.0% (Silverman & Barasch)	No red counts or daily streaks; neutral copy; amnesty and fresh start; forecasts always paired with a fix
6	Notifications are too weak, or too many	Tiimo: "quick ding isn't enough"; Routinery: "inconsistent notifications"	Distinct Watch haptics per event type; AlarmKit only for user-chosen critical moments; a daily budget that learns which prompts get ignored; rolling 48-hour window under iOS's 64-notification cap
7	Constant UI change disorients users	Tiimo: "layout and features change OFTEN which can be unsettling" (App Store AU)	A stable navigation contract; new features opt-in; TestFlight beta group for changes
8	Dependence instead of skill	CBT's benefit comes from skills the person uses themselves (Safren 2010)	A "plan with me" mode that builds the day together; routine graduation; weekly review teaches the pattern, not just the result

13.2 Medication, regulation and privacy

#	Risk	Evidence	Mitigation
9	The window is wrong for the individual	57% of 1,327 lisdexamfetamine users report 7 h or less vs a 14 h label (ADxS)	User's own estimate first; check-ins; uncertainty shown; high-stakes blocks (exams) never lean on an early estimate
10	Reclassified as a medical device	TGA is reviewing its mental-health exclusion (62% say it's no longer appropriate); MHRA names "time of treatment" as device territory	The red lines in section 8; a claims register reviewed before every release; a lawyer review before launch; watch the TGA consultation
11	A privacy breach or a perceived one	Health providers were the top breach-notifying sector in 2025 (225 of 1,205, OAIC); OAIC ruled against tracking pixels on health data in June 2026	No ad or analytics SDKs anywhere health data appears; on-device processing; end-to-end encrypted sync; minimal server; a written breach plan
12	Association with stimulant misuse	Done Global's founder sentenced to 72 months in July 2026 (DOJ); 52% of top TikTok ADHD videos misleading (Yeung 2022)	No "study hack" framing; no sharing features; clinician-reviewed marketing claims; creator partnerships with a disclosure policy
13	Irregular dosing and shortages break the plan	71.5% of US adults on stimulants had trouble filling scripts (CDC)	"No meds today", late and partial days re-plan instantly; shortage mode; plans stay usable unmedicated

13.3 Platform and technical

#	Risk	Evidence	Mitigation
14	iOS won't run the app in the background	~30 s refresh at the system's discretion; silent push throttled	Event-driven upkeep (11.2)

#	Risk	Evidence	Mitigation
15	Calendar sync breaks, duplicates or double-books	Motion: "double-booked meetings", "calendar sync issues"; Reclaim: 26 reviews cite poor calendar integration	App-owned calendar; UUIDs in the URL field; idempotent imports; reconcile on every wake; never edit third-party events without a confirm; QA across iCloud, Google-via-iOS and Exchange
16	Users refuse full calendar access	Write-only access can't read even the app's own events (TN3153)	Explain before the system prompt; a degraded mode where users mark busy times; let users exclude sensitive calendars
17	Apple Watch app is flaky	Tiimo users logged out on the Watch mid-routine	Watch works offline with no login dependency; logs write locally and sync later
18	Learning looks dumb on sparse data	~20% of check-ins go unanswered; early estimates are noisy	Defaults and shrinkage; honest "early estimate" labels; passive signals fill gaps
19	AI hallucinates or is hijacked by a screenshot	"The AI hallucinates a lot" (Motion on Trustpilot); prompt injection via images (OWASP)	Fixed output schemas; the solver checks feasibility; previews before writes; parsed content never touches medication
20	Older iPhones get a worse app	Apple Intelligence needs iPhone 15 Pro or later	Every AI feature has a grammar or manual path

13.4 Business

#	Risk	Evidence	Mitigation
21	Subscription resentment ("ADHD tax")	Motion billing complaints (Trustpilot 3.7 from 578); accidental annual purchases on Inflow; users praise Structured's scholarship	Free core; trial-end reminder 2 days before; one-tap cancel instructions; student and hardship pricing; optional lifetime

#	Risk	Evidence	Mitigation
22	Trial ends before learning shows value	"One week is not enough" (Inflow review); 17 to 32-day trials convert 42.5% vs 25.5% for trials under 4 days (RevenueCat 2026)	A 21 to 30-day trial; show "what I've learned about your week" before the paywall
23	AI features churn and cost	AI apps retain 36% worse over 12 months (RevenueCat 2026); third-party AI ~US$0.35 to $1.05 per active user a month	Position as a planner, not a chatbot; on-device and Private Cloud Compute by default
24	A bigger player copies the idea	Tiimo (2025 iPhone App of the Year, 50K+ paying subscribers by Aug 2024) plans energy-level learning; four medication-curve trackers could add scheduling; Apple's Health redesign is in beta	Ship the learned window plus scheduling first; compound the personal data moat; stay Apple-native and featurable; full data export so users trust the switch
25	Australia alone is too small	AU rating counts run 7 to 9% of US counts for Tiimo and Structured; ~600K Australians dispensed ADHD medication in 2023–24 (AIHW)	Build US and UK localisation (spelling, formulation names such as Elvanse, term structures) from v1; use Australia as the beta market
26	Scope outruns one builder	The inventory plus these additions is well over a hundred features	The phased roadmap in section 17; a legal entity from day one; cut anything that doesn't serve the core loop

14. Competitive landscape and differentiation

Two product classes exist and nothing found joins them. ADHD medication trackers (Theraview, Pillr, Medly, FocusCurve) draw onset and wear-off curves but schedule nothing; planners and AI schedulers schedule everything but ignore medication. Two 2026 vendor round-ups reached the same conclusion (Saner.ai, 44 apps; Carly). The gap is a personally learned focus window that drives the schedule, plus a term/holiday toggle nobody offers.

14.1 Who's out there (September 2026)

App	What it is	Price	Rating	Medication-aware	Learns from the user	Main complaint
Tiimo	ADHD visual planner with AI co-planner, Watch, Reminders sync; 2025 iPhone App of the Year	Free; Pro A$9 to A$66	AU 4.42 (1,588); US 4.59 (20,231)	No	Partly; pattern and energy learning listed as future	Frequent layout changes; Watch logouts; weak alerts; rigid minute blocks
Structured	Visual day timeline; claims 500K+ Pro subscribers	Free; Pro monthly, yearly, lifetime; scholarship	AU 4.74 (14,796); US 4.79 (166,591)	No	No evidence	None visible
Inflow	CBT programme, AI companion, body doubling; bought by Cerebral March 2026	US$29.99 a month	AU 4.24 (417); US 4.37 (5,621)	No ("not a medical device")	No evidence	Cost; 1-week trial too short
Motion	AI auto-scheduler, now project management	US$19 to $29 a seat a month (a review says US$49)	Trustpilot 3.7 (578)	No	Reschedules, no personal learning shown	Over-packed days; billing; "AI hallucinates"
Reclaim	AI habit and focus-time defender	Free; US$10 to $22 a month	G2 4.8 (144)	No	Adaptive habit placement	No Apple Calendar; no mobile app; clutter
Sunsama	Guided daily planning	US$20 to $25 a month	Not captured	No	Logs planned vs actual, doesn't predict	Setup; iCloud not listed
FlowSavvy	Auto-scheduler with iCloud sync and "scheduling hours profiles"	Free; Pro US$10 to $14	Not captured	No	No	—

App	What it is	Price	Rating	Medication-aware	Learns from the user	Main complaint
Lifestack	Energy-zone scheduling from Apple Health, WHOOP, Oura	Not shown	Not captured	No	Yes, energy from wearables	—
Goblin Tools	AI task breakdown micro-tools	Web free; app A$2.99	AU 4.52 (226)	No	No	"Hit-or-miss"
Focus Bear (AU)	Routines plus distraction blocking	Free; Pro US$9.99 a month	Not captured	No	Partial	—
FocusCurve	Formulation-aware medication and caffeine curves, on-device, refill alerts	Subscription or lifetime	Not captured	Yes, population curves	No	Doesn't schedule tasks
Pillr	Stimulant window (opening, active, fading), wear-off alerts	Free; Premium US$14.99	AU 5.0 (5)	Yes	No	Tiny user base
Theraview	Onset and wear-off visualiser with manual "metabolic rate"	Free	US 4.90 (328)	Yes, 9 medicines	Manual only	Last iOS update August 2023
Apple Health Medications	Built-in log and reminders; readable by other apps since iOS 26	Free	—	Logs only	No	No effect modelling

Ratings from Apple's lookup service, 28 September 2026 (AU). Clockwise, a decade-old AI calendar, shut down on 27 March 2026 with a week's notice (The Stack): a reminder that users need export and trust.

14.2 What makes this app different

1. **The focus window drives the schedule.** Trackers draw population curves; this app learns the person's own window from their check-ins and puts their chosen work inside it.
2. **Personal duration multipliers with confidence.** No surveyed scheduler documents learning how long this person actually takes.
3. **Lifestyles.** A one-tap term/holiday switch with separately learned routines. The closest thing is FlowSavvy's paid scheduling profiles, which don't learn.
4. **Log once.** Reading doses users already log in Apple Health removes double entry. No competitor was seen doing this.
5. **Apple-native and Watch-first.** Reclaim has no Apple Calendar support and Sunsama doesn't list iCloud, which leaves room for an app built around Calendar, Reminders, the Watch and Live Activities.
6. **Designed for lapses.** Amnesty, baseline days and shame-free re-entry are core, not settings.
7. **Privacy by architecture.** Health context stays on the device or in Apple's Private Cloud Compute.

14.3 Market size

Market	Figure	Source
Australia, people dispensed ADHD medication	~600,000 in 2023–24 (22 per 1,000 people, up from 8 in 2018–19)	AIHW
Australia, adults among new patients	49.9% in 2022–23, up from 32.8% in 2019–20	PBS DUSC
US adults with current ADHD	15.5M; 33.4% took stimulants in the past year	CDC 2024
UK adults with ADHD	~1.9M (charity estimate)	ADHD UK

A 1% paid conversion of Australia's medicated population is about 6,000 subscribers, which supports planning US and UK releases early.

14.4 Pricing and distribution

- **Free core:** dose logging with the double-dose guard, the focus-window display, manual planning, amnesty and re-entry. Charging for the safety features would read as an ADHD tax.

- **Paid:** auto-scheduling, learning, Lifestyles, forecasting, roster import. Annual plan, a student price and a hardship scholarship (Structured's scholarship is praised in reviews).
- **Trial:** 21 to 30 days, because the learning needs about two weeks to show; a reminder 2 days before it ends.
- **Optional lifetime** tier with a cap on third-party AI use.
- **Apple takes 15%** under the Small Business Program below US$1M a year (Apple).
- **Channels:** Apple editorial (Tiimo's path); the shareable log export as a reason clinicians mention the app; university disability and wellbeing services; creator partnerships with clinician-reviewed claims.

15. Privacy, security and data governance

Treat everything medication-related, including the inferred focus window, as health information. Collect it only with specific opt-in consent, keep it on the device where possible, sync it end-to-end encrypted outside iCloud, and never let it reach advertising, analytics or third-party AI. Building to the strictest rule in every launch market from day one (Washington's opt-in, GDPR explicit consent, Apple 5.1.2(i)) is simpler than retrofitting.

15.1 Data classes

Class	Examples	Where it lives	Leaves the device?	Consent
Medication	Medicines, dose logs, boosters, supply, check-ins, the focus-window model	Encrypted on-device store	Only to Apple's Private Cloud Compute for AI features (stateless, nothing kept), as end-to-end encrypted sync between the user's own devices, or in an export they start	Specific opt-in at first medication setup

Class	Examples	Where it lives	Leaves the device?	Consent
Health-sourced	Sleep, heart rate, workouts, cycle, State of Mind	Read on demand; only derived features kept	Never raw; at most a non-health summary ("a lighter day")	Apple's per-type prompt plus an in-app explanation first
Calendar and reminders	Events, reminders, the app's ID map	Apple's stores plus the local map	Via the user's own iCloud, Google or Exchange; calendars are never end-to-end encrypted	Full calendar access, explained first
Planning data	Tasks, ledger, rules, preferences, lifestyles	On-device plus encrypted sync	Encrypted sync only	Account terms
Text and screenshots for AI	"Lunch at 1", roster images	Processed on-device or in Private Cloud Compute	Third-party AI only if opted in, de-identified, no health content	Explicit opt-in naming the provider and country
Location	Saved places, geofences	On-device	No	Asked when the first errand is added
Product analytics	Screen counts, crashes	Aggregated	Yes, but never from medication or health screens, and crash logs scrubbed of medicine names	Opt-in

15.2 Sync without breaking App Store rule 5.1.3(ii)

Apple says apps "may not store personal health information in iCloud", which rules out plain CloudKit sync of dose logs.

Option	How	Verdict
Local-only	iPhone holds everything; the Watch talks to the phone directly	MVP: simplest and safest, but no Mac or iPad
Own end-to-end encrypted sync	Devices encrypt with keys only they hold; a minimal server (Australian region) stores ciphertext it can't read	v1: enables Mac and iPad without the app ever seeing health data
CloudKit with encrypted fields	Apple's sync with field-level encryption	Still "in iCloud"; only with Apple's written confirmation **[confirm]**

15.3 Legal obligations, Australia first

- **Health information is sensitive information** under the Privacy Act. Consent must be informed, voluntary, current and specific; the OAIC presumes capacity from age 15 (OAIC Chapter B).
- **Probably a "health service"**, because it records health, so the small-business exemption likely doesn't apply: full Australian Privacy Principles from day one **[confirm]** (OAIC).
- **Sending data to US AI providers is a cross-border disclosure** unless the app keeps effective control, and the app stays accountable for what the provider does (APP 8). The OAIC recommends not entering sensitive information into public AI tools at all (OAIC AI guidance).
- **Automated decisions:** from 10 December 2026, privacy policies must describe the personal information used in automated decision-making that significantly affects people (Privacy and Other Legislation Amendment Act 2024). A calendar planner likely falls below that bar, but disclosing it costs nothing.
- **Breaches:** notify the OAIC and affected users when serious harm is likely. 2025 set a record of 1,205 notifications, with health service providers the top sector (OAIC).
- **US:** the amended FTC Health Breach Notification Rule (from 29 July 2024) covers health apps and counts unauthorised sharing as a breach (Federal Register). Washington's My Health My Data Act requires opt-in consent and allows private lawsuits (WA AG).
- **EU and UK:** explicit consent for health data (GDPR Art 9) and a data protection impact assessment before launch.

15.4 AI data flow

Tier	Where it runs	What it may see	Retention
0. Grammar	On-device code	Anything	None
1. Apple Foundation Models	On-device	Anything, including medication context	None
2. Apple Private Cloud Compute	Apple servers, stateless	Health-adjacent parsing and long inputs	Kept only until the response returns (Apple Security)
3. Third-party model (Claude, OpenAI)	Provider's cloud, paid API tier	Opt-in, de-identified productivity text only; never medication, check-ins or Health data	Up to ~30 days by default; zero-retention by agreement; no training on API data (OpenAI, Anthropic)

Gemini is excluded: its API terms ban medical advice and apps likely to be used by under-18s (Gemini terms). Don't promise "your data stays in Australia" unless a contract guarantees where inference runs **[confirm]**.

15.5 Security basics

- Optional Face ID lock on the app and on medication screens.
- iOS file protection on the store, with widget and extension data in a shared container readable after first unlock.
- No medicine names in notification text, calendar titles or Live Activities in discreet mode.
- No third-party SDKs on any screen that shows health data; crash reports scrubbed.
- An independent security test before launch, repeated yearly.
- A written breach-response plan covering the Australian scheme and the US 60-day rule.

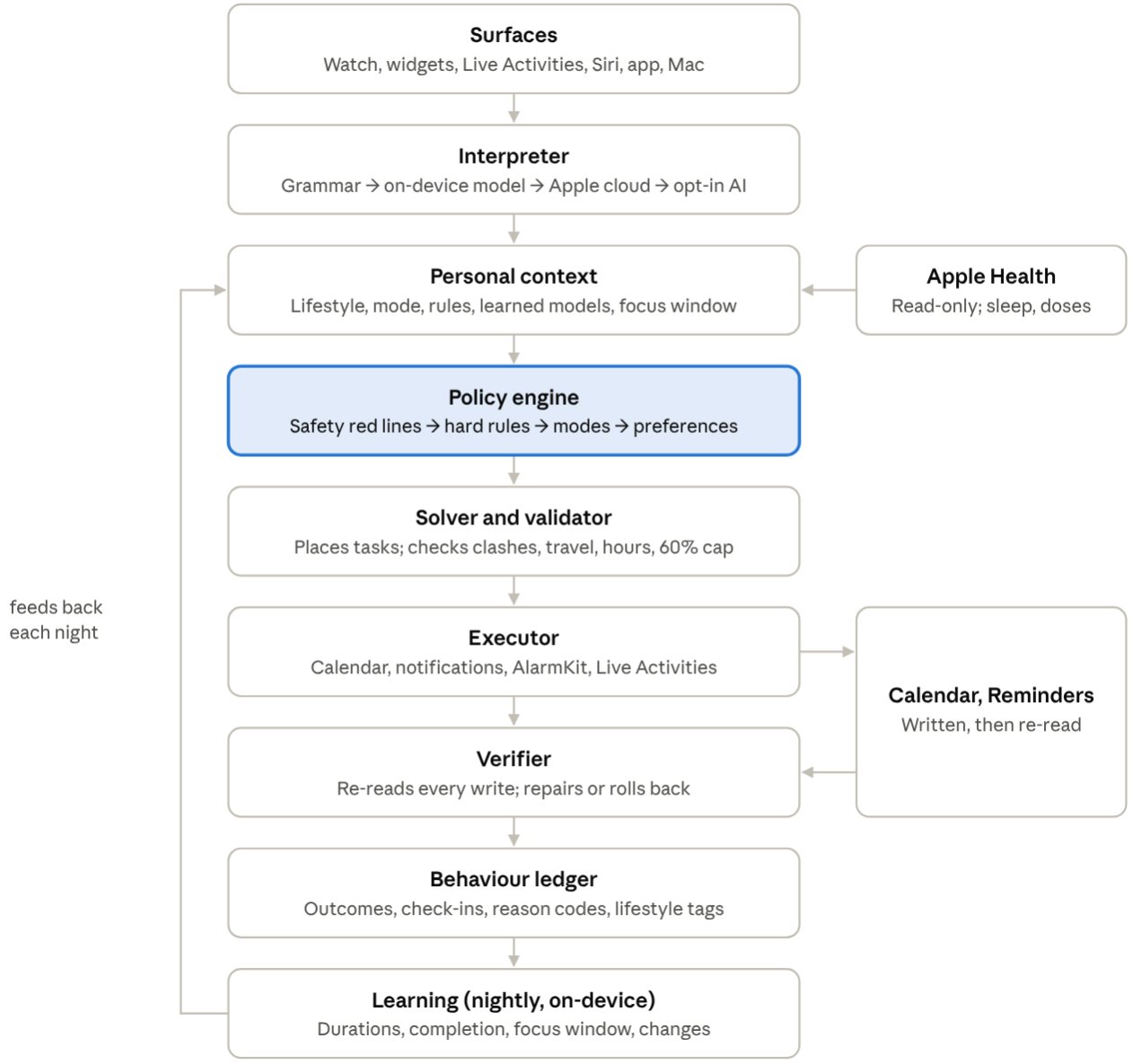
15.6 User controls

- A "What leaves this phone" page listing every destination, what goes there and why.
- Per-category switches (the original inventory's section 57), with medication and Health data locked to on-device and Private Cloud Compute.
- Export everything (JSON and CSV), with a warning that the file contains health data.
- Delete everything, in the app, including server copies and backups.
- Withdraw any consent from the same screen where it was given.

16. Architecture and data model (revised)

The original chain stays, with four changes: a safety layer at the top of the policy engine, lifestyles and modes inside the personal context, event-driven upkeep instead of a background daemon, and an AI interpreter that tries free, private tiers before any cloud model.

Rules and a solver decide; the model only interprets; learning loops back



Revised architecture · 9 stages, 2 Apple stores, nightly learning loop

Apple Health feeds context read-only. Calendar and Reminders are written by the executor and read back by the verifier before anything counts as done.

16.1 Canonical objects

The original list stays. These are added or changed:

Object	Key fields	Notes
`Medication`	Name, class (stimulant or not), prescribed plan text, cut-off time, minimum re-log interval	User-entered; never edited by the app
`Product`	Brand, formulation, strength, curve prior, stored label note	A brand switch creates a new product
`DoseEvent`	Product, time taken, time logged, source (app, Watch, Health import), food flag, booster flag	Health imports keyed by Health's ID for de-duplication
`CheckIn`	Time, answer (yes, partly, no), optional energy, optional mood	Feeds the focus-window model
`FocusWindowModel`	Per product: grid posterior summary, evidence count, stated estimate, manual override	Shown only as "your focus pattern"
`Lifestyle`	Name, active flag, per-lifestyle statistics snapshot, episode dates, scheduled dates	Dormant snapshots never update
`Mode`	Rule overlay, start and end dates, stacking order	Exam, Sick day, Travel
`Rule`	Level (safety, hard, temporary, learned, heuristic), scope (all or one lifestyle), source text, confirmation	Safety rules are compiled in, not user-editable
`LearnedPattern`	Statement, evidence count, interval, status (proposed, approved, rejected, cooling off)	30-day cool-off after a rejection
`Task`	Adds focus demand (deep, medium, light, body), lifestyle scope, reason codes on lapses	Most fields inferred
`SupplyRecord`	Quantity on hand, repeats left, script expiry, eScript token, pharmacy notes	Neutral supply view only

Object	Key fields	Notes
`ExternalObjectLink`	App UUID, external ID, calendar ID, occurrence date, reminder ID, sync state	UUID also written into the event's URL field
`PlanRevision`	Before, after, actor, rule, reason, undo token	Powers undo, restore and "why"
`ConsentRecord`	Category, scope, provider, country, version, time	Every consent can be withdrawn where it was given

16.2 Tech stack

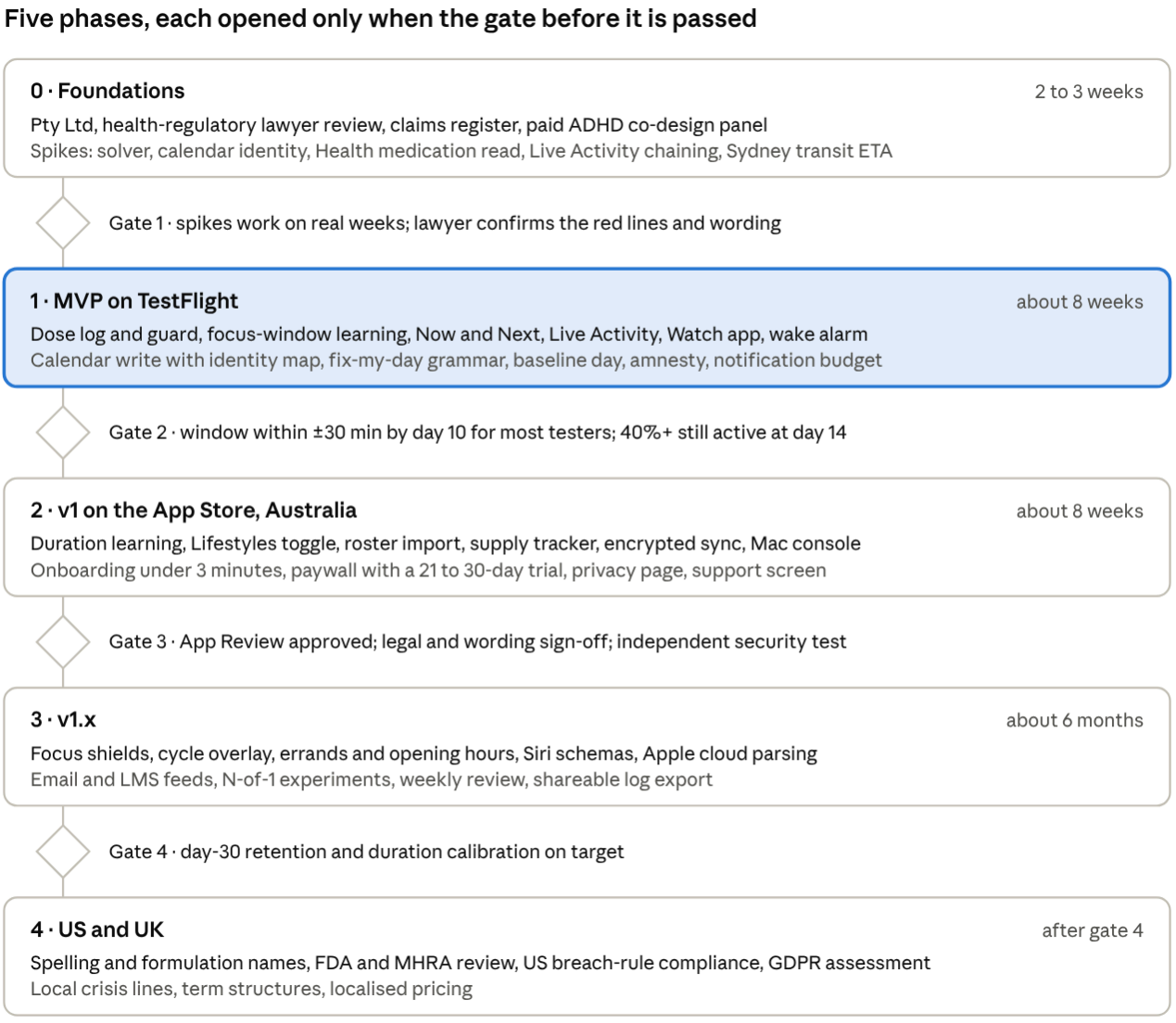
Layer	Choice
UI	SwiftUI multiplatform (iPhone, Watch, iPad, Mac); WidgetKit; ActivityKit; `AssistiveAccess` scene
Local store	SwiftData, local configuration (no CloudKit for health data); shared App Group container for widgets and extensions
Learning	Plain Swift: conjugate Bayesian updates, grid posteriors, change detection; runs on launch and nightly
Solver	A small constraint solver on-device; prove the choice in a 2-week spike against real week-long plans
AI	FoundationModels (on-device and Private Cloud Compute), Vision for OCR; optional third-party model behind the same `LanguageModel` interface
Apple services	EventKit, HealthKit (read), App Intents, AlarmKit, UserNotifications, CoreLocation, MapKit
Server (minimal)	End-to-end encrypted sync relay that stores only ciphertext, plus a silent-push backstop; Australian region

16.3 Precedence, restated

Safety red lines (compiled in, section 8) → user hard rules → active mode → active lifestyle's rules → approved learned preferences → optimisation heuristics. A learned preference can never override a hard rule, and nothing can override a safety rule, including the user's own chat instructions ("just tell me when to take my booster" gets the fixed referral).

17. Build roadmap, MVP scope and validation

Build the core loop first: log a dose, learn the window, fill it with chosen work, survive a bad day. Everything else waits until testers show that loop works and keeps them coming back. Durations assume one person building full time; your inventory's exam period (9 to 27 November) suggests Phase 0 spikes in October and the MVP build over the summer break.



Roadmap · 5 phases, 4 gates; durations are estimates for one full-time builder

17.1 What the MVP includes, and what it leaves out

In the MVP	Deliberately left out
Medication setup with the user's own estimate; one-tap log from Watch, widget, Control Center and Siri; double-dose guard; undo	Supply tracker, shortage mode, cycle overlay
Focus-window learning with 3 to 5 check-ins a day, tapering	Heart-rate signal, objective probe
Today screen with Now and Next, Live Activity, Watch complication, app-owned wake alarm	Mac and iPad apps, sync
Tasks with duration and deadline; solver places deep work in the window; app-owned calendar; identity map	Roster import, travel, opening hours, errands
Fix my day via fixed phrases; baseline day; amnesty; welcome-back flow; notification budget	Free-form chat, third-party AI
Calm mode, setup presets, support screen, consent flows	Spark themes, Screen Time shields, Lifestyles

Lifestyles move to v1 because they need at least one learned lifestyle to switch from; testers running over the summer break are the ideal first Lifestyle 2 users.

17.2 Gate criteria (proposed targets)

Gate	Pass when	Measured by
1	The solver plans a real week in under a second on an older iPhone; calendar identity survives calendar moves, recurring edits and Exchange; Health medication read works; written legal advice on the red lines	Spike reports, lawyer's letter
2	For at least 70% of medicated testers, the focus window's 80% range is within ±30 minutes by day 10; at least 40% still active at day 14; no medication advice in a 500-prompt red-team test	In-app logs, red-team script

Gate	Pass when	Measured by
3	App Review approves HealthKit use and the third-party AI disclosure; privacy policy, consent flows and the support screen pass review; no high-severity security findings	App Store Connect, pen-test report
4	Day-30 retention of at least 20% (the mental-health app median is 3.3%); P50 duration predictions hit 40 to 60% and P80 hit 70 to 90%; refunds below the category's 4.71%	Cohort analytics, calibration report, App Store data

17.3 How to validate

- **Pilot cohort:** 20 to 40 adults with diagnosed ADHD, most on stimulants, a mix of students and workers, recruited through ADHD communities. If results are to be published, get university ethics approval first.
- **Co-design panel:** 8 to 12 paid ADHD users who see every major screen before it ships (HCI reviews single out missing co-design as the field's main weakness).
- **Built-in comparisons:** alternate pairs of days with the learned duration multiplier on or off for consenting testers, to measure whether learning actually improves adherence (section 9.7).
- **Weekly qualitative check:** three questions, never more: what helped, what annoyed you, what did you ignore.

17.4 If time runs short, cut in this order

1. Screen Time shields and Siri schemas.
2. Errands, opening hours and travel.
3. Mac console (the Watch and iPhone carry the core).
4. Roster import (manual shift entry still works).

Never cut the double-dose guard, undo, the baseline day, amnesty or the safety rules.

18. Open decisions for you

Ten decisions shape everything else; each has a recommendation, but they're yours to make.

#	Decision	Options	Recommendation	Why
1	Minimum age	18+, or 16+	18+ at launch	Gemini bars under-18 apps; Anthropic requires age checks for minors; the Children's Online Privacy Code lands by 10 Dec 2026
2	What the window is called	Focus window, energy window, medication window	"Focus window"	Keeps it a self-reported pattern, not a drug-effect prediction (section 8)
3	Is your own regime the default?	Your rules as defaults for everyone, or as one template	One template among several	Most users don't share "no gym on work days"; configuration presets do the job
4	Pricing	Free core plus paid learning; hard paywall with trial	Free core plus paid learning, 21 to 30-day trial, student price	Safety features shouldn't cost money; learning needs weeks to show value
5	Sync at launch	Local-only; own encrypted sync; CloudKit	Local-only for MVP, own encrypted sync for v1	App Store 5.1.3(ii) makes CloudKit risky for health data
6	Third-party AI	None; opt-in for non-health text; default on	None at launch; opt-in later	On-device and Private Cloud Compute cover the core; fewer consent and cross-border issues
7	Import doses from Apple Health	Yes (optional), no	Yes, optional	Removes double logging for people already using Apple's Medications

#	Decision	Options	Recommendation	Why
8	Health data at launch	Sleep and workouts; add heart rate and cycle later	Sleep and workouts only	Minimum data for the core loop; fewer permission prompts
9	First audience	University students; working adults; both	University students first, adults a close second	Term/holiday Lifestyles, assessments and rosters fit students best, and it's the network you can reach
10	Build route	Solo, with a co-founder, with contractors for compliance	Solo build plus paid legal and security review	The code is buildable by one person; the regulatory and security sign-offs aren't

Some items need outside help regardless: a health-regulatory and privacy lawyer (the exact TGA exclusion wording, whether ADHD counts as "serious", the Australian data-residency position), Apple's own reading of rule 5.1.3(ii), and an independent security test.

19. Sources

Research run on 28 September 2026. Some primary sources (PubMed, several journals, most TGA pages, Reddit) blocked automated access; where a secondary summary was used, the text says so. Full notes with every source, organised by topic, are saved to this Project.

ADHD struggles and evidence

- Marx et al. 2021, time perception meta-analysis (ADHD Evidence summary)
- Fuermaier et al. 2013, prospective memory and planning, PLOS ONE
- Beheshti et al. 2020, emotional dysregulation meta-analysis, BMC Psychiatry
- Chen, Meng & Nie 2026, adults with ADHD and productivity tools, arXiv/CSCW
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- ADHD UK medication crisis report
- The Conversation on Australian ADHD medication shortages, 2025
- Kasturiratna et al. 2025, digital interventions meta-analysis
- Baumel et al. 2019, mental-health app retention (summary)

Interventions and design

- Safren et al. 2010, CBT for adult ADHD, JAMA
- Solanto et al. 2010, meta-cognitive therapy, AJP
- AADPA Australian guideline factsheet
- Barkley fact sheet on executive function
- HeartSteps micro-randomised trial, Annals of Behavioral Medicine 2019
- Nahum-Shani et al. 2018, JITAI framework
- Silverman & Barasch, broken streaks
- Inflow feasibility study, PLOS Digital Health 2022
- W3C cognitive accessibility guidance

Medication

- Australian Prescriber, adult ADHD pharmacology, Feb 2026
- FDA Concerta label; FDA Ritalin LA label; FDA Vyvanse label
- Medsafe Concerta data sheet
- AU consumer medicine information: Vyvanse, Aspen Dexamfetamine
- Wigal et al. 2010, lisdexamfetamine adult workplace study
- LDX vs d-amphetamine hysteresis, Frontiers in Pharmacology 2017
- ADxS real-world duration survey (secondary)
- NSW ACI ADHD in general practice FAQ
- ADHD Mums shortage tracker, Aug 2026

Learning and lifestyles

- Murphy, conjugate Bayesian analysis of the Gaussian
- Liao et al. 2019, Personalized HeartSteps
- Buehler, Griffin & Ross 1994, planning fallacy
- Adams & MacKay 2007, online change-point detection
- Bei et al. 2014, adolescent sleep in school vs vacation
- AHRQ N-of-1 trials user's guide

Apple platform

- WWDC25 HealthKit Medications API

- TN3153 EventKit access changes
- Live Activities documentation
- TN3193 Foundation Models context window
- WWDC26 Apple Intelligence guide
- App Review Guidelines

Regulation and privacy

- TGA software guidance (mirror, Aug 2025); TGA software and AI reforms, July 2026
- FDA enforcement-discretion examples
- MHRA stand-alone software guidance
- MDCG 2019-11 rev.1
- OAIC APP 8 guidelines; OAIC AI guidance
- FTC Health Breach Notification Rule
- DOJ on Done Global sentencing, July 2026
- Gemini API additional terms

Market and competitors

- AIHW ADHD medications dispensed
- CDC MMWR 2024, US adult ADHD
- RevenueCat subscription benchmarks 2026
- Tiimo funding and traction
- FocusCurve
- Saner.ai 2026 review of 44 ADHD apps